

Succinct Betti-2 Positivity and Its Complexity

Cauchy-Based Evaluation, $\oplus \mathbf{P}$ -Completeness on a Promise Family, Deterministic SAT Reductions, and Unconditional Circuit Lower Bounds on Explicit Subfamilies

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Abstract

We study the decision problem (Π_{β_2}) : given a succinct (local-oracle) description (D) of a bounded-degree simplicial (2)-complex (K_D) , decide whether the second Betti number $(\beta_2(K_D; \mathbb{F}_2))$ is positive. The central theme is the interaction between (i) topological invariants of low-dimensional complexes and (ii) computational complexity under succinct representations, with implications for the landscape surrounding (P) versus (NP) .

We present two complementary reduction pipelines. The first pipeline defines a Cauchy-based evaluation problem (SCE-Dec) over $(\mathbb{F}_{2^k})$, proves that SCE-Dec can be expressed as an $(\mathbb{F}_2)$ -linear form in evaluator bits, and encodes this linear form as a bounded-occurrence $(\mathbb{F}_2)$ -linear system. We then convert such systems into CW (2)-complexes and further into bounded-degree simplicial (2)-complexes via a star-triangulation procedure. This yields a mapping (ProbeBit) from SCE-Dec instances to local-oracle descriptions (D) such that $(\beta_2(K_D) > 0)$ if and only if the target SCE-Dec bit is (1). As complexity consequences, we obtain $(\oplus \mathbf{P})$ -completeness of (Π_{β_2}) restricted to the promise family $(\text{Im}(\text{ProbeBit}))$, and a one-sided randomized many-one reduction $(\text{SAT} \leq_{rp} \Pi_{\beta_2})$ on the same promise family using the Valiant–Vazirani isolation lemma [1]. We also prove an unconditional deterministic evaluation-local query lower bound $(q \geq N)$ for computing SCE-Dec.

The second pipeline gives a deterministic “witness-expansion” construction that maps a Boolean formula (ϕ) to a disjoint union (K_ϕ) of constant-size sphere/disk gadgets indexed by assignments. This yields a parsimonious equality $(\beta_2(K_\phi; \mathbb{F}_2) = \#\text{SAT}(\phi))$, hence deterministic $(\text{SAT} \leq_m \Pi_{\beta_2})$ and $(\#\text{SAT} \leq_m \text{Compute-}\beta_2)$ (function hardness). Finally, we provide unconditional nonuniform lower bounds against $(\mathbf{AC}^0)$, De Morgan formulas, and $(\mathbf{AC}^0[p])$ on explicit parity-based subfamilies via projection reductions from (PARITY) (standard results (see, e.g., [3, 4, 7, 5, 6])).

We explicitly separate validated statements from steps requiring either standard citations or remaining formalization (oracle-interface encoding details and a triangulated-disk homeomorphism lemma). No unconditional separation such as $(P \neq NP)$ is claimed.

1 Introduction

1.1 Motivation and context

The question (P) versus (NP) sits at the center of theoretical computer science. One broad strategy toward understanding the landscape around (P) and (NP) is to analyze natural computational problems that arise outside of “synthetic” encodings, and to determine how their structural content (algebraic, geometric, topological) interacts with algorithmic and lower-bound methods. Topological invariants, and Betti numbers in particular, provide canonical examples: they are intrinsic,

stable under homeomorphisms, and encode global structure in a way that is often difficult to infer from local information.

This paper concerns the complexity of deciding whether the second Betti number (β_2) of a simplicial (2)-complex is positive, when the complex is given succinctly. The decision problem we study is:

Given a local-oracle (succinct) description (D) of a bounded-degree simplicial (2)-complex (K_D), decide whether ($\beta_2(K_D; \mathbb{F}_2) > 0$).

The bounded-degree condition is essential: it keeps local neighborhoods constant-size, enabling local-oracle access to faces, edges, and incident structure. However, even under bounded degree, a succinctly represented complex can have exponentially many simplices relative to the input length, making it nontrivial to relate “local query access” to global invariants such as (β_2).

While the main topic is not a proof of ($P \neq NP$), the results are relevant to (P) versus (NP) in three ways:

1. Hardness and completeness phenomena for (Π_{β_2}) (both under promise restrictions and in unrestricted deterministic NP-hardness via explicit gadget families) illustrate how global topological invariants naturally encode counting and satisfiability.
2. Query lower bounds in oracle models demonstrate information-theoretic barriers for evaluation tasks intimately related to the reductions.
3. Unconditional circuit lower bounds on explicit subfamilies give a clean, “white-box” separation of (Π_{β_2}) from low-depth circuit classes on carefully chosen inputs, complementing the more algebraic and topological reductions.

1.2 Two reduction pipelines

We develop two distinct pipelines, each with different strengths and technical requirements.

Pipeline A (Cauchy/SCE $\rightarrow$ linear systems $\rightarrow$ topology: the ProbeBit route). We define a succinct evaluation problem SCE-Dec based on multiplying a vector ($x(X) \in \mathbb{F}_{2^k}^N$), generated by an evaluator circuit (X), by an explicit Cauchy matrix ($C_N \in \mathbb{F}_{2^k}^{N \times N}$). The target is a single output bit of the form

$$\text{SCE}_{N,i,t}(X) = \pi_t((C_N x(X))_i) \in \mathbb{F}_2.$$

We show that $\text{SCE}_{N,i,t}(X)$ is an ($\mathbb{F}_2$)-linear form in the evaluator output bits ($x_{j,\ell}$) with coefficients ($m_{j,\ell}^{(N,i,t)} \in \mathbb{F}_2$). We encode this linear form as a bounded-occurrence ($\mathbb{F}_2$)-linear system $\text{SysBit}(N, i, t, b, X)$ such that the system has a nonzero solution if and only if $\text{SCE}_{N,i,t}(X) = b$. Next, we build a CW (2)-complex $K^{\text{cw}}(\text{Sys})$ whose second homology (H_2) is isomorphic to the solution space $\text{Sol}(\text{Sys})$. Finally, we convert the CW complex into a bounded-degree simplicial (2)-complex $K(\text{Sys})$ by triangulating each (2)-cell via a star-triangulation; in the intended form, this triangulation preserves homology. The composition yields a mapping (ProbeBit) from SCE-Dec instances to local-oracle descriptions of bounded-degree simplicial (2)-complexes such that

$$\beta_2(K_{\text{ProbeBit}(N,i,t,b,X)}; \mathbb{F}_2) > 0 \iff \text{SCE}_{N,i,t}(X) = b.$$

Pipeline B (Witness-expansion route: deterministic SAT and #SAT). Independently, we define a deterministic construction mapping a Boolean formula (ϕ) to a disjoint union

$$K_\phi = \bigsqcup_{x \in \{0,1\}^n} C_x,$$

where each component $\mathcal{C}_x$ is a constant-size simplicial (2)-complex that is a “sphere gadget” if $\phi(x) = 1$ and a “disk gadget” if $\phi(x) = 0$. Over $(\mathbb{F}_2)$, the sphere gadget has $(\beta_2 = 1)$ and the disk gadget has $(\beta_2 = 0)$. By additivity of homology over disjoint unions, we obtain

$$\beta_2(K_\phi; \mathbb{F}_2) = \#\text{SAT}(\phi),$$

which immediately implies deterministic $(\text{SAT} \leq_m \Pi_{\beta_2})$ and parsimonious $(\#\text{SAT} \leq_m \text{Compute-}\beta_2)$.

1.3 Main results and careful scope statements

We state high-level theorems here and provide full proofs in Parts 2–5. Statements that rely on standard external results are explicitly labeled. Where a step is nontrivial but standard, we cite a primary source; all remaining technical steps are proved in the indicated sections/appendices.

Theorem 1.1 (ProbeBit correctness; proof in Parts 2–4; one topological lemma in Section 11). *There exists an explicit mapping (ProbeBit) taking an SCE-Dec instance $((N, i, t, b, X))$ to a local-oracle description (D) of a bounded-degree simplicial (2)-complex (K_D) such that*

$$\beta_2(K_D; \mathbb{F}_2) > 0 \iff \text{SCE}_{N,i,t}(X) = b.$$

Proof status: the algebraic, gadget, and CW-homology steps are proved in Parts 2–3; the simplicialization and homology-preservation step uses a triangulated-disk homeomorphism lemma proved in Section 11.

Theorem 1.2 $((\oplus \mathbf{P})$ -completeness on a promise family; proof in Part 4). *Let $\mathcal{I}_{\text{Probe}} := \text{Im}(\text{ProbeBit})$ be the promise family of local-oracle descriptions produced by (ProbeBit). Then (Π_{β_2}) restricted to $\mathcal{I}_{\text{Probe}}$ is $(\oplus \mathbf{P})$ -complete under deterministic many-one reductions.*

Scope note: This is a promise statement: hardness and membership are shown only for inputs guaranteed to lie in $\mathcal{I}_{\text{Probe}}$.

Theorem 1.3 (One-sided randomized SAT reduction on the ProbeBit promise family; standard external lemma). *There is a one-sided randomized many-one reduction*

$$\text{SAT} \leq_{rp} \Pi_{\beta_2} \upharpoonright_{\mathcal{I}_{\text{Probe}}}.$$

Proof uses: Valiant–Vazirani isolation lemma [1].

Scope note: Again a promise statement.

Theorem 1.4 (Evaluation-local query lower bound; unconditional; proof in Part 4). *Fix (N, i, t) . Any deterministic evaluation-local algorithm that computes $\text{SCE}_{N,i,t}(X)$ for all evaluator circuits (X) must query $(X(j))$ for at least (N) distinct indices $(j \in [N])$ in the worst case.*

Theorem 1.5 (Witness-expansion parsimonious equality; deterministic; proof in Part 5). *There exists a deterministic mapping $(\phi \mapsto D_\phi)$ producing a local-oracle description (D_ϕ) of a bounded-degree simplicial (2)-complex (K_{D_ϕ}) such that*

$$\beta_2(K_{D_\phi}; \mathbb{F}_2) = \#\text{SAT}(\phi).$$

Consequently, $(\text{SAT} \leq_m \Pi_{\beta_2})$ and $(\#\text{SAT} \leq_m \text{Compute-}\beta_2)$. Proof status: topological facts about the constant-size gadgets are proved directly (finite chain computation), and the disjoint-union additivity is proved in Part 5.

Theorem 1.6 (Unconditional circuit lower bounds on explicit subfamilies; proof in Part 5; standard external results). *There exists an explicit subfamily $\mathcal{F}$ of instances of (Π_{β_2}) such that $\Pi_{\beta_2} \upharpoonright_{\mathcal{F}}$ computes (PARITY) under a projection reduction. Hence:*

- $(\Pi_{\beta_2} \notin \mathbf{AC}^0)$ (nonuniform),
- (Π_{β_2}) requires quadratic-size De Morgan formulas on $(\mathcal{F})$, and
- $(\Pi_{\beta_2} \notin \mathbf{AC}^0[p])$ for odd primes (p) ,

using standard results about (PARITY) [3, 4, 7, 5, 6].

Important clarification: These are unconditional lower bounds, but they are proved via an explicit hard subfamily $(\mathcal{F})$. The implication to the global language is by restriction: if the full language were in a circuit class, then its restriction to $(\mathcal{F})$ would be as well.

1.4 What this paper does not claim

- We do not claim an unconditional separation such as $(P \neq NP)$.
- Several results are explicitly formulated as promise results (notably $(\oplus \mathbf{P})$ -completeness and $(\text{SAT} \leq_{rp})$ on $(\text{Im}(\text{ProbeBit}))$).
- We keep a strict separation between:
 - deterministic NP-hardness via witness-expansion (unrestricted), and
 - promise-family completeness via ProbeBit.

1.5 Organization of the paper (across five parts)

- Part 1 (this part): definitions, models, and problem statements.
- Part 2: SCE-Dec definition; linearization into XOR masks; bounded-occurrence gadgets; construction of SysBit.
- Part 3: CW encoding of linear systems; theorem $(H_2 \cong \text{Sol})$.
- Part 4: simplicialization and homology preservation; ProbeBit mapping; $(\oplus \mathbf{P})$ -completeness on promise family; $\text{SAT} (\leq_{rp})$; evaluation-local lower bound.
- Part 5: witness-expansion (deterministic SAT and #SAT); unconditional circuit lower bounds on explicit parity-based subfamilies; discussion, limitations, appendices, and bibliography.

2 Notation and Preliminaries

2.1 Basic notation and conventions

- For $(m \in \mathbb{N})$, we write

$$[m] := \{0, 1, \dots, m-1\},$$

i.e., we use 0-indexing throughout.

- All vector spaces are over $(\mathbb{F}_2)$ unless explicitly stated otherwise. We write XOR as $(\oplus)$. For vectors $(u, v \in \mathbb{F}_2^d)$, $(u \oplus v)$ denotes coordinate-wise XOR.

- For a finite set (S) , we use $(\mathbb{F}_2^S)$ to denote the $(\mathbb{F}_2)$ -vector space of formal $(\mathbb{F}_2)$ -linear combinations of elements of (S) , equivalently functions $(S \rightarrow \mathbb{F}_2)$.
- Unless stated otherwise, “polynomial time” and “polynomial size” are with respect to the length of the explicit input bitstring (e.g., a Boolean formula (ϕ) or an oracle/circuit description (D)).
- Degree bounds as constants. When we say “bounded degree” for a simplicial complex in the local-oracle model, the bound (Δ) is treated as a fixed absolute constant that is part of the model, not part of the input length.

2.2 Finite fields of characteristic two

We require arithmetic in $(\mathbb{F}_{2^k})$ to define the Cauchy-based evaluation problem. To avoid symbol collisions, we reserve $(\mathbb{K})$ for the finite field and (K) for a simplicial complex.

Definition 2.1 (Finite field model $(\mathbb{K} = \mathbb{F}_{2^k})$ via a polynomial basis). *Fix $(k \in \mathbb{N})$. Let $(p_k(z) \in \mathbb{F}_2[z])$ be an irreducible polynomial of degree (k) . Define*

$$\mathbb{K} := \mathbb{F}_{2^k} \cong \mathbb{F}_2[z]/(p_k(z)).$$

Let $\alpha := z \bmod p_k(z)$. We use the basis

$$\mathcal{B} := \{1, \alpha, \alpha^2, \dots, \alpha^{k-1}\}$$

to identify $(\mathbb{K})$ with $(\mathbb{F}_2^k)$ as a vector space.

Every element $(u \in \mathbb{K})$ has a unique coordinate representation

$$u = \sum_{\ell=0}^{k-1} u_\ell \alpha^\ell, \quad u_\ell \in \mathbb{F}_2.$$

Definition 2.2 (Coordinate projections). *For each $(t \in [k])$, define $\pi_t : \mathbb{K} \rightarrow \mathbb{F}_2$ by*

$$\pi_t \left(\sum_{\ell=0}^{k-1} u_\ell \alpha^\ell \right) := u_t.$$

Definition 2.3 (Bitstring encoding map). *Given a bitstring $(z = (z_0, \dots, z_{k-1}) \in \mathbb{F}_2^k)$, define*

$$\text{enc}(z) := \sum_{\ell=0}^{k-1} z_\ell \alpha^\ell \in \mathbb{K}.$$

When convenient, we view integers $(r \in [2^k])$ as bitstrings $(z \in \mathbb{F}_2^k)$ via binary expansion and then as field elements via enc .

Remark 2.4 (Boolean circuit realizability of basic field operations). *Addition in $(\mathbb{K})$ is coordinate-wise XOR under $(\mathcal{B})$. Multiplication can be implemented by polynomial multiplication followed by reduction modulo (p_k) . Inversion can be implemented by an extended Euclidean algorithm in $(\mathbb{F}_2[z])$ modulo (p_k) .*

Proof status: The concrete circuit bounds (size $\text{poly}(k)$) are proved in Appendix B. This point is needed only to justify succinctness and time bounds in later sections.

2.3 Simplicial (2)-complexes and $(\mathbb{F}_2)$ -homology

We work with simplicial complexes of dimension at most (2) and with coefficients in $(\mathbb{F}_2)$.

Definition 2.5 (Simplicial (2)-complex). *A simplicial (2)-complex is a triple $(K = (V, E, T))$ where*

- (V) is a finite set of vertices,
- $(E \subseteq \binom{V}{2})$ is a set of (unordered) edges,
- $(T \subseteq \binom{V}{3})$ is a set of (unordered) triangles,

such that the closure condition holds: if $\{u, v, w\} \in T$, then $\{u, v\}, \{u, w\}, \{v, w\} \in E$. We assume (K) has no simplices of dimension ≥ 3 .

Definition 2.6 (Chain groups and boundary maps over $(\mathbb{F}_2)$). *Let $(K = (V, E, T))$ be a simplicial (2)-complex. Define chain groups*

$$C_2(K) := \mathbb{F}_2^T, \quad C_1(K) := \mathbb{F}_2^E, \quad C_0(K) := \mathbb{F}_2^V.$$

Define boundary maps $\partial_2 : C_2 \rightarrow C_1$ and $\partial_1 : C_1 \rightarrow C_0$ on basis simplices by

$$\partial_2(\{u, v, w\}) := \{u, v\} \oplus \{u, w\} \oplus \{v, w\},$$

$$\partial_1(\{u, v\}) := u \oplus v,$$

and extend $(\mathbb{F}_2)$ -linearly. Since (K) has no (3)-simplices, $C_3(K) = 0$ and $\text{im}(\partial_3) = \{0\}$.

Definition 2.7 (Second homology and second Betti number). *The second homology group over $(\mathbb{F}_2)$ is*

$$H_2(K; \mathbb{F}_2) := \ker(\partial_2) \subseteq C_2(K),$$

and the second Betti number is

$$\beta_2(K; \mathbb{F}_2) := \dim_{\mathbb{F}_2} H_2(K; \mathbb{F}_2).$$

Remark 2.8 (Why $(\mathbb{F}_2)$ coefficients). *All constructions in this paper are tailored to $(\mathbb{F}_2)$: (i) linear systems are over $(\mathbb{F}_2)$, (ii) parity encodings are natural, and (iii) the simplicial/cellular boundary computations are most transparent. Some gadget-based arguments extend to other fields, but we restrict to $(\mathbb{F}_2)$ for consistency and to minimize ancillary technicalities.*

2.4 Bounded degree and local-oracle descriptions

The central object of study is a simplicial (2)-complex given succinctly via a local-oracle description.

Definition 2.9 (Bounded-degree simplicial (2)-complex). *Let $(\Delta \in \mathbb{N})$. A simplicial (2)-complex $(K = (V, E, T))$ is (Δ) -bounded-degree if every vertex $(v \in V)$ is incident to at most (Δ) edges and at most (Δ) triangles.*

We treat (Δ) as a fixed constant throughout.

Definition 2.10 (Local-oracle description of a bounded-degree simplicial (2)-complex). *A local-oracle description (D) of a (Δ) -bounded-degree simplicial (2)-complex consists of:*

1. *Three nonnegative integers (n_V, n_E, n_T) (in binary), intended as sizes of (V, E, T) .*

2. Boolean circuits implementing the following functions:

- Edge endpoints oracle

$$\text{End}_D : [n_E] \rightarrow [n_V] \times [n_V],$$

where $\text{End}_D(e) = (u, v)$ returns the endpoints of edge (e) .

- Triangle vertices oracle

$$\text{Vert}_D : [n_T] \rightarrow [n_V] \times [n_V] \times [n_V],$$

where $\text{Vert}_D(\tau) = (u, v, w)$ returns the vertices of triangle (τ) .

- Incidence listing oracles

$$\text{IncE}_D : [n_V] \times [\Delta] \rightarrow [n_E] \cup \{\perp\}, \quad \text{IncT}_D : [n_V] \times [\Delta] \rightarrow [n_T] \cup \{\perp\},$$

where $\text{IncE}_D(v, \ell)$ returns the ℓ -th edge incident to (v) (or $\perp$), and $\text{IncT}_D(v, \ell)$ returns the ℓ -th triangle incident to (v) (or $\perp$).

3. A promise that these circuits describe a valid (Δ) -bounded-degree simplicial (2) -complex $(K_D = (V_D, E_D, T_D))$ with $(V_D = [n_V])$, edges (E_D) induced by End_D , triangles (T_D) induced by Vert_D , satisfying:

- closure (every triangle's edges exist in (E_D)),
- consistency with incidence lists,
- bounded-degree by (Δ) .

Remark 2.11 (Why incidence lists are included). Without IncE_D and IncT_D , recovering all incident edges/triangles around a vertex from End_D and Vert_D alone may require $\Omega(n_E)$ or $\Omega(n_T)$ work. The bounded-degree promise allows constant-length incidence lists, making “local access” genuinely local.

Definition 2.12 (Semantic validity predicate $\text{Valid}(D)$). Fix the global constant degree bound $\Delta \geq 1$ from the local-oracle model. Let D be a local-oracle description with parameters (n_V, n_E, n_T) and oracles $\text{End}_D, \text{Vert}_D, \text{IncE}_D, \text{IncT}_D$ of the types in Definition 2.10.

We say $\text{Valid}(D)$ holds if all of the following semantic conditions hold:

1. **Range/type constraints.** For every $e \in [n_E]$, $\text{End}_D(e) = (u, v)$ satisfies $u, v \in [n_V]$. For every $\tau \in [n_T]$, $\text{Vert}_D(\tau) = (a, b, c)$ satisfies $a, b, c \in [n_V]$. For every $(v, \ell) \in [n_V] \times [\Delta]$, $\text{IncE}_D(v, \ell) \in [n_E] \cup \{\perp\}$ and $\text{IncT}_D(v, \ell) \in [n_T] \cup \{\perp\}$.
2. **No loops / distinct triangle vertices.** For every $e \in [n_E]$, if $\text{End}_D(e) = (u, v)$ then $u \neq v$. For every $\tau \in [n_T]$, if $\text{Vert}_D(\tau) = (a, b, c)$ then a, b, c are pairwise distinct.
3. **No duplicate simplices (injective up to permutation).** For all distinct edges $e \neq e' \in [n_E]$, writing $\text{End}_D(e) = (u, v)$ and $\text{End}_D(e') = (u', v')$, we have $\{u, v\} \neq \{u', v'\}$. Likewise, for all distinct triangles $\tau \neq \tau' \in [n_T]$, writing $\text{Vert}_D(\tau) = (a, b, c)$ and $\text{Vert}_D(\tau') = (a', b', c')$, we have $\{a, b, c\} \neq \{a', b', c'\}$.
4. **Incidence list distinctness.** For every $v \in [n_V]$, the non- $\perp$ values among $\{\text{IncE}_D(v, \ell) : \ell \in [\Delta]\}$ are pairwise distinct. Likewise, the non- $\perp$ values among $\{\text{IncT}_D(v, \ell) : \ell \in [\Delta]\}$ are pairwise distinct.

5. **Incidence soundness (edges).** For every $v \in [n_V]$ and $\ell \in [\Delta]$, if $\text{IncE}_D(v, \ell) = e \in [n_E]$ and $\text{End}_D(e) = (u, w)$, then $v \in \{u, w\}$.
6. **Incidence completeness (edges).** For every $e \in [n_E]$ with $\text{End}_D(e) = (u, w)$, there exist $\ell_u, \ell_w \in [\Delta]$ such that $\text{IncE}_D(u, \ell_u) = e$ and $\text{IncE}_D(w, \ell_w) = e$.
7. **Incidence soundness (triangles).** For every $v \in [n_V]$ and $\ell \in [\Delta]$, if $\text{IncT}_D(v, \ell) = \tau \in [n_T]$ and $\text{Vert}_D(\tau) = (a, b, c)$, then $v \in \{a, b, c\}$.
8. **Incidence completeness (triangles).** For every $\tau \in [n_T]$ with $\text{Vert}_D(\tau) = (a, b, c)$, there exist $\ell_a, \ell_b, \ell_c \in [\Delta]$ such that $\text{IncT}_D(a, \ell_a) = \tau$, $\text{IncT}_D(b, \ell_b) = \tau$, and $\text{IncT}_D(c, \ell_c) = \tau$.
9. **Closure (2-simplices have boundary edges).** For every $\tau \in [n_T]$ with $\text{Vert}_D(\tau) = (a, b, c)$, there exist $e_{ab}, e_{ac}, e_{bc} \in [n_E]$ such that $\text{End}_D(e_{ab})$ has endpoints $\{a, b\}$, $\text{End}_D(e_{ac})$ has endpoints $\{a, c\}$, and $\text{End}_D(e_{bc})$ has endpoints $\{b, c\}$ (unordered endpoints).

Remark (scope). $\text{Valid}(D)$ is a semantic promise predicate; we do not claim it is decidable in polynomial time.

Definition 2.13 (Size and query time in the local-oracle model). The description size ($|D|$) is the total bit-length of the encoding of (n_V, n_E, n_T) plus the encodings of the four circuits $\text{End}_D, \text{Vert}_D, \text{IncE}_D, \text{IncT}_D$. A query to (D) is one evaluation of one oracle circuit on a valid input. We say (D) is efficiently queriable if each oracle evaluation runs in time $\text{poly}(|D|)$ on a standard RAM model.

Fixed bit-level encoding. We fix once and for all an explicit, injective bit-level encoding function $\text{Enc}(\cdot) : \text{Desc} \rightarrow \{0, 1\}^*$ for local-oracle descriptions, together with a deterministic total parser $\text{Parse} : \{0, 1\}^* \rightarrow (\text{Desc} \cup \{\perp\})$, as specified in Appendix C. We write $n := |\text{Enc}(D)|$ for the input length and measure all running times in $\text{poly}(n)$. When we refer to the “description length” $|D|$, we mean $|D| := |\text{Enc}(D)|$ under this fixed convention.

2.5 Complexity classes and reductions used later

We recall the basic complexity notions used in Parts 4–5.

Definition 2.14 ($(\oplus \mathbf{P})$). A language $(L \subseteq \{0, 1\}^*)$ is in $(\oplus \mathbf{P})$ if there exists a nondeterministic polynomial-time Turing machine (M) such that for all (x) ,

$$x \in L \iff \#\text{acc}_M(x) \equiv 1 \pmod{2},$$

where $\#\text{acc}_M(x)$ is the number of accepting computation paths of (M) on input (x) .

Definition 2.15 ($(\oplus \text{SAT})$). $(\oplus \text{SAT})$ is the decision problem: given a Boolean formula (ϕ) , decide whether $\#\text{SAT}(\phi)$ is odd.

Standard result. $(\oplus \text{SAT})$ is $(\oplus \mathbf{P})$ -complete under deterministic many-one reductions [2].

Definition 2.16 (Deterministic many-one reduction). For languages $(A, B \subseteq \{0, 1\}^*)$, we write $(A \leq_m B)$ if there is a polynomial-time computable function (f) such that $(x \in A \iff f(x) \in B)$.

Definition 2.17 (One-sided randomized many-one reduction). We write $(A \leq_{rp} B)$ if there exists a randomized polynomial-time function (f) and a polynomial (p) such that:

- If $(x \in A)$, then $\Pr[f(x) \in B] \geq 1/p(|x|)$;

- If $(x \notin A)$, then $\Pr[f(x) \in B] = 0$.

Definition 2.18 (Promise problems and restriction to a family). *Given a language/problem (B) and a subset $(\mathcal{I})$ of inputs, we write $(B \upharpoonright_{\mathcal{I}})$ for the restriction of (B) to inputs in $(\mathcal{I})$. When we say a hardness/completeness statement holds “on $(\mathcal{I})$,” we mean it is a promise statement: inputs are guaranteed to lie in $(\mathcal{I})$.*

3 Computational Problems Studied

3.1 Betti-2 positivity and related function problems

Definition 3.1 (Betti-2 positivity problem (Π_{β_2})). *The decision problem (Π_{β_2}) takes as input a local-oracle description (D) promised to describe a valid (Δ) -bounded-degree simplicial (2) -complex (K_D) , and outputs*

$$\Pi_{\beta_2}(D) := \begin{cases} 1 & \text{if } \beta_2(K_D; \mathbb{F}_2) > 0, \\ 0 & \text{if } \beta_2(K_D; \mathbb{F}_2) = 0. \end{cases}$$

Definition 3.2 (Total β_2 -positivity language). *Fix the encoding Enc and total parser Parse from Appendix C, and the semantic predicate $\text{Valid}(\cdot)$ from Definition 2.12. Define the total language*

$$L_{\beta_2} := \left\{ x \in \{0, 1\}^* : \text{Parse}(x) = D \neq \perp, \text{Valid}(D), \beta_2(K_D; \mathbb{F}_2) > 0 \right\}.$$

Equivalently, if $\text{Parse}(x) = \perp$ or $\text{Parse}(x) = D$ but $\neg \text{Valid}(D)$, then $x \notin L_{\beta_2}$ (invalid inputs reject).

Remark 3.3 (Relation to the promise problem). *For every valid local-oracle description D (i.e. $\text{Valid}(D)$ holds), we have*

$$\text{Enc}(D) \in L_{\beta_2} \iff \Pi_{\beta_2}(D) = 1.$$

Definition 3.4 (Compute- (β_2) as a function problem). *The function problem $\text{Compute-}\beta_2$ takes as input (D) and outputs the integer $\beta_2(K_D; \mathbb{F}_2)$ (in binary).*

Remark 3.5 (Decision vs function hardness). *A parsimonious reduction $(\#\text{SAT}(\phi) \mapsto \beta_2(K_\phi))$ yields $(\#\mathbf{P})$ -hardness for $\text{Compute-}\beta_2$ and NP -hardness for (Π_{β_2}) , but does not imply membership of (Π_{β_2}) in (NP) . In the succinct setting, membership questions can be nontrivial because witnesses (e.g., a nonzero (2) -cycle) might be exponentially large in the description length.*

3.2 A baseline polynomial-space upper bound

Theorem 3.6 (Polynomial-space upper bound for Π_{β_2}). *There is a deterministic Turing machine that, on input a (succinct) local-oracle description D (Definition 2.10) promised to satisfy $\text{Valid}(D)$ (Definition 2.12) and to be efficiently queriable (Definition 2.13), decides whether $\beta_2(K_D; \mathbb{F}_2) > 0$ using space $\text{poly}(|D|)$. In particular, $\Pi_{\beta_2} \in \mathbf{PSPACE}$.*

Moreover, under the fixed parsing convention of Definition 3.2, the total language L_{β_2} is also decidable in $\mathbf{PSPACE}$.

Proof. Let $m := n_E$ and $n := n_T$ for the given description D .

Step 0: the easy case $n > m$. The boundary map $\partial_2 : C_2(K_D) \rightarrow C_1(K_D)$ is an $\mathbb{F}_2$ -linear map from an n -dimensional space to an m -dimensional space. Hence $\text{rank}(\partial_2) \leq m$ and

$$\dim \ker(\partial_2) = n - \text{rank}(\partial_2) \geq n - m > 0.$$

Thus if $n_T > n_E$ then $\Pi_{\beta_2}(D) = 1$.

Assume henceforth $n \leq m$.

Step 1: access to the boundary matrix entrywise. Index $C_2(K_D) = \mathbb{F}_2^{T_D}$ by triangle indices $\tau \in [n_T]$ and $C_1(K_D) = \mathbb{F}_2^{E_D}$ by edge indices $e \in [n_E]$. Let $A \in \mathbb{F}_2^{m \times n}$ denote the matrix of ∂_2 in these standard bases, i.e.,

$$A_{e,\tau} = 1 \iff \text{edge } e \text{ appears in } \partial_2(\tau).$$

Given (e, τ) , we can compute $A_{e,\tau}$ in time $\text{poly}(|D|)$ using only $O(\Delta)$ oracle queries: compute $\text{Vert}_D(\tau) = (a, b, c)$; then find the (unique) edge indices $e_{ab}, e_{ac}, e_{bc} \in [n_E]$ with endpoints $\{a, b\}, \{a, c\}, \{b, c\}$ by scanning the constant-size incidence list $\text{IncE}_D(a, \cdot)$ and checking endpoints via End_D . Existence (and uniqueness) of these boundary edges is guaranteed by $\text{Valid}(D)$. Finally, output $A_{e,\tau} = 1$ iff $e \in \{e_{ab}, e_{ac}, e_{bc}\}$.

Step 2: reduce $\beta_2 > 0$ to a rank test. Because K_D has no 3-simplices, $H_2(K_D; \mathbb{F}_2) = \ker(\partial_2)$ (Definitions 2.6 and 2.7). Therefore

$$\beta_2(K_D; \mathbb{F}_2) > 0 \iff \ker(A) \neq \{0\} \iff \text{rank}(A) < n.$$

Step 3: computing $\text{rank}(A)$ in polynomial space. It is a standard fact that matrix rank over a fixed finite field (in particular over $\mathbb{F}_2$) admits logspace-uniform polylogarithmic-depth circuit families (e.g., via parallel Gaussian elimination), so there is a logspace-uniform circuit family $\{\mathcal{C}_{m,n}\}$ of depth $O(\log^2(m+n))$ that outputs $\text{rank}(A)$ given the mn input bits of A .

We evaluate $\mathcal{C}_{m,n}$ *on the fly* without materializing all of its (possibly exponentially many) gates. Logspace-uniformity means that, given (m, n) and a gate index, a deterministic procedure using space $O(\log(mn)) = \text{poly}(|D|)$ can compute the gate type and the indices of its children. A depth-first evaluation of a circuit of depth $O(\log^2(m+n)) = \text{poly}(|D|)$ uses only $\text{poly}(|D|)$ space for the recursion stack and for storing intermediate field elements.

Whenever the evaluation reaches an input gate corresponding to an entry (e, τ) , we compute $A_{e,\tau}$ by the oracle procedure from Step 1, still within $\text{poly}(|D|)$ space. Hence we obtain $\text{rank}(A)$ using $\text{poly}(|D|)$ space, and we accept iff $\text{rank}(A) < n$.

Total language. To decide L_{β_2} (Definition 3.2), first run $\text{Parse}(x)$. Then verify $\text{Valid}(D)$ by iterating over all indices in $[n_V], [n_E], [n_T]$ using counters of $O(\log n_V + \log n_E + \log n_T) = \text{poly}(|x|)$ bits and checking the finitely many semantic clauses. Finally apply the above polynomial-space procedure for $\beta_2 > 0$. $\square$

Remark 3.7 (Witness-size blowup vs. polynomial-space membership). *The naive NP witness for $\beta_2 > 0$ is an explicit nonzero 2-cycle $z \in \ker(\partial_2) \subseteq \mathbb{F}_2^{n_T}$, for example by listing all triangles of z with coefficient 1. In the local-oracle model, n_T is given in binary and may be $2^{\Theta(|D|)}$, so an explicit z can be exponentially long in the description length.*

Moreover, bounded-degree does not force the existence of a sparse 2-cycle: there are bounded-degree 2-complexes (e.g., large connected triangulations of closed surfaces with no 3-simplices) in which every nonzero 2-cycle necessarily has support $\Omega(n_T)$. This explains why NP-membership is nontrivial in the succinct setting, while Theorem 3.6 shows a robust polynomial-space upper bound.

3.3 A succinct Cauchy-based evaluation problem (SCE-Dec)

To connect (Π_{β_2}) to parity counting, we introduce an auxiliary problem based on multiplication by a Cauchy matrix over $(\mathbb{F}_{2^k})$. Full details appear in Part 2; we state the definition now since it motivates subsequent reductions.

We will set $k := \lceil \log_2(4N) \rceil$, ensuring $2^k \geq 4N$. We define an explicit $(N \times N)$ Cauchy matrix (C_N) over $(\mathbb{K} = \mathbb{F}_{2^k})$ using two disjoint sets of field elements $\{a_0, \dots, a_{N-1}\}$ and $\{b_0, \dots, b_{N-1}\}$ (precise construction in Part 2).

An evaluator (X) is a Boolean circuit that, on input $(j \in [N])$, outputs (k) bits interpreted as an element $(x_j(X) \in \mathbb{K})$. This defines a vector $(x(X) \in \mathbb{K}^N)$.

Definition 3.8 (SCE bit). *Given integers $(N \geq 1)$, $(i \in [N])$, $(t \in [k])$, and an evaluator (X) , define*

$$\text{SCE}_{N,i,t}(X) := \pi_t((C_N x(X))_i) \in \mathbb{F}_2.$$

Definition 3.9 (SCE decision language SCE-Dec). *Define*

$$\text{SCE-Dec} := \{(N, i, t, b, X) : \text{SCE}_{N,i,t}(X) = b\},$$

where $b \in \mathbb{F}_2$.

Remark 3.10 (Why SCE-Dec is a natural intermediary). *The Cauchy matrix gives a dense linear operator, and the output bit $\text{SCE}_{N,i,t}(X)$ can be expressed as a parity (XOR) of evaluator output bits with explicit coefficients. This parity structure is the bridge that allows encoding into bounded-occurrence $(\mathbb{F}_2)$ -linear systems, and then into (H_2) of a (2) -complex.*

3.4 Promise families and scope of hardness statements

Two distinct families of instances arise:

1. ProbeBit family $(\mathcal{I}_{\text{Probe}})$: instances (D) that are guaranteed to be of the form $(D = \text{ProbeBit}(N, i, t, b, X))$. Many parity-counting completeness results are established on this family (Part 4).
2. Witness-expansion family $(\mathcal{I}_{\text{WE}})$: instances (D_ϕ) produced by the deterministic witness-expansion construction mapping formulas (ϕ) to complexes (K_ϕ) . Deterministic $(\text{SAT} \leq_m \Pi_{\beta_2})$ is shown on this family (Part 5), and because the map is deterministic and polynomial-time, this yields NP-hardness for the unrestricted decision problem (Π_{β_2}) .

We will keep these families separate. In particular:

- $(\oplus \mathbf{P})$ -completeness is proved for $(\Pi_{\beta_2} \upharpoonright_{\mathcal{I}_{\text{Probe}}})$, not for all inputs.
- Deterministic NP-hardness is proved via witness-expansion, without using ProbeBit's promise restriction.

4 Background and Related Work (brief)

This section is intentionally brief; we cite a few standard primary sources for the external results we use.

- Succinct representations. Many graph and combinatorial problems become substantially harder under succinct encodings (e.g., circuit-encoded adjacency predicates). Our local-oracle model for bounded-degree simplicial complexes is an instance of this general paradigm.
- Homology computation. Computing Betti numbers is a classical problem in computational topology. Complexity varies sharply with dimension, coefficient field, and representation format (explicit vs succinct). We focus on bounded-degree simplicial (2) -complexes under local-oracle access, and on the specific decision predicate $(\beta_2 > 0)$ over $(\mathbb{F}_2)$.

- Parity and counting classes. The class $(\oplus \mathbf{P})$ and $(\oplus \text{SAT})$ are standard objects in complexity theory [2], with $(\oplus \text{SAT})$ being $(\oplus \mathbf{P})$ -complete.
- Valiant–Vazirani isolation. The Valiant–Vazirani lemma [1] provides a randomized reduction from SAT to UniqueSAT. In this paper it is used to obtain a one-sided randomized reduction from SAT to (Π_{β_2}) on a promise family.
- Circuit lower bounds. Lower bounds for (PARITY) against $(\mathbf{AC}^0)$, De Morgan formulas, and $(\mathbf{AC}^0[p])$ are classical [3, 4, 7, 5, 6]. We use these results in a “white-box” manner by exhibiting explicit parity-encoding subfamilies of (Π_{β_2}) .

5 Status Markers and Resolved Technical Items

Earlier drafts listed several items as “proof-deferred” or uncited. In this version:

- The topological triangulated-disk boundary-fixing homeomorphism lemma is proved in Section 11 (Lemma 11.8).
- The fixed bit-level encoding of oracle circuits is specified in Appendix C.
- The external results we use are cited explicitly (e.g., Valiant–Vazirani isolation [1], $\oplus \text{SAT}$ completeness [2], and standard PARITY lower bounds [3, 4, 7, 5, 6]).

Part II: Field-Linearity, Bit-Linear Forms for SCE, and the Bounded-Occurrence System SysBit

6 Field-Linearity and Bit-Linear Forms for SCE

This section gives the algebraic core underlying the later topological encodings: a target output bit of a succinct Cauchy evaluation can be written as an explicit $\mathbb{F}_2$ -linear form in evaluator output bits. We also record a basic “nonzero-row” property that will later support an information-theoretic lower bound.

Throughout, we use the finite-field representation from Definition 2.1: for each k , we fix an irreducible polynomial $p_k \in \mathbb{F}_2[z]$ of degree k , set $\mathbb{K} = \mathbb{F}_2[z]/(p_k)$, write $\alpha = z \bmod p_k$, and use the basis $\mathcal{B} = \{1, \alpha, \dots, \alpha^{k-1}\}$.

6.1 Cauchy matrix over $\mathbb{F}_{2^k}$

We now define the explicit Cauchy matrix used in the Succinct Cauchy Evaluation (SCE) problem.

Definition 6.1 (Field size for parameter N , evaluation points, and Cauchy matrix). *Fix an integer $N \geq 2$. Set*

$$k := \lceil \log_2(4N) \rceil, \quad \mathbb{K} := \mathbb{F}_{2^k},$$

represented using the polynomial-basis model from Definition 2.1.

Define the field elements

$$a_i := \text{enc}(i) \in \mathbb{K} \quad \text{for } i \in [N], \quad b_j := \text{enc}(N+j) \in \mathbb{K} \quad \text{for } j \in [N],$$

where $\text{enc}(\cdot)$ is the bitstring-to-field encoding from Definition 2.3 (using the k -bit binary expansion of an integer in $[2^k]$).

Define the Cauchy matrix

$$C_N \in \mathbb{K}^{N \times N} \quad \text{by} \quad (C_N)_{i,j} := \frac{1}{a_i - b_j}.$$

(Over characteristic two, subtraction equals addition; we retain $a_i - b_j$ for readability.)

Lemma 6.2 (Disjointness of $\{a_i\}$ and $\{b_j\}$). *For all $i, j \in [N]$, $a_i \neq b_j$. Consequently, $(C_N)_{i,j}$ is well-defined and nonzero.*

Proof. By definition $k = \lceil \log_2(4N) \rceil$, hence $2^k \geq 4N > 2N$. Therefore all integers in $\{0, 1, \dots, 2N-1\}$ are distinct elements of $[2^k]$. The encoding $\text{enc} : [2^k] \rightarrow \mathbb{K}$ is injective (Definition 2.3). Thus $\text{enc}(i) \neq \text{enc}(N+j)$ for all $i, j \in [N]$, i.e., $a_i \neq b_j$. Hence $a_i - b_j \neq 0$ in $\mathbb{K}$, so the inverse exists and is nonzero. $\square$

6.2 Evaluators and the SCE target bit

We formalize the “evaluator” notion used informally in Part I.

Definition 6.3 (Evaluator and implicit vector $x(X)$). *Fix $N \geq 2$ and $k = \lceil \log_2(4N) \rceil$. An evaluator is a Boolean circuit*

$$X : [N] \rightarrow \{0, 1\}^k.$$

For each $j \in [N]$, write

$$X(j) = (x_{j,0}, x_{j,1}, \dots, x_{j,k-1}) \in \mathbb{F}_2^k,$$

and interpret it as a field element

$$x_j(X) := \sum_{\ell=0}^{k-1} x_{j,\ell} \alpha^\ell \in \mathbb{K}.$$

This defines an implicit vector $x(X) = (x_0(X), \dots, x_{N-1}(X)) \in \mathbb{K}^N$.

Recall from Definition 3.8 that for indices $i \in [N]$ and $t \in [k]$,

$$\text{SCE}_{N,i,t}(X) := \pi_t((C_N x(X))_i) \in \mathbb{F}_2.$$

6.3 Multiplication by a fixed field element is $\mathbb{F}_2$ -linear

Under a fixed basis, multiplication by a fixed $\kappa \in \mathbb{K}$ is an $\mathbb{F}_2$ -linear operator and hence corresponds to a $k \times k$ matrix over $\mathbb{F}_2$.

Definition 6.4 (Multiplication operator). *For $\kappa \in \mathbb{K}$, define*

$$L_\kappa : \mathbb{K} \rightarrow \mathbb{K}, \quad L_\kappa(u) = \kappa u.$$

Lemma 6.5 ($\mathbb{F}_2$ -linearity and the coordinate matrix $M(\kappa)$). *For every $\kappa \in \mathbb{K}$, the map L_κ is $\mathbb{F}_2$ -linear. Consequently, there exists a unique matrix*

$$M(\kappa) \in \mathbb{F}_2^{k \times k}$$

such that for every $u = \sum_{\ell=0}^{k-1} u_\ell \alpha^\ell \in \mathbb{K}$,

$$\kappa u = \sum_{t=0}^{k-1} \left(\bigoplus_{\ell=0}^{k-1} M(\kappa)_{t,\ell} u_\ell \right) \alpha^t.$$

Proof. Linearity: for $u, v \in \mathbb{K}$ and $c \in \mathbb{F}_2 = \{0, 1\}$, $L_\kappa(u \oplus v) = \kappa(u + v) = \kappa u + \kappa v = L_\kappa(u) \oplus L_\kappa(v)$, and $L_\kappa(cu) = c(\kappa u)$ holds because c is central and $c \in \{0, 1\}$. Thus L_κ is $\mathbb{F}_2$ -linear.

Existence/uniqueness of $M(\kappa)$: the basis $\mathcal{B}$ identifies $\mathbb{K}$ with $\mathbb{F}_2^k$ as vector spaces. Any $\mathbb{F}_2$ -linear map $\mathbb{K} \rightarrow \mathbb{K}$ is determined uniquely by its values on the basis vectors α^ℓ for $\ell \in [k]$, which become the columns of the matrix $M(\kappa)$. $\square$

Lemma 6.6 (Invertibility for $\kappa \neq 0$). *If $\kappa \in \mathbb{K} \setminus \{0\}$, then $M(\kappa)$ is invertible as a matrix over $\mathbb{F}_2$. In particular, every row and every column of $M(\kappa)$ is nonzero.*

Proof. If $\kappa \neq 0$, multiplication by κ is a bijection $\mathbb{K} \rightarrow \mathbb{K}$ (since $\mathbb{K}$ is a field). Therefore the $\mathbb{F}_2$ -linear map L_κ is invertible, hence its matrix representation $M(\kappa)$ is invertible over $\mathbb{F}_2$.

If a row of $M(\kappa)$ were all zeros, then the corresponding output coordinate $\pi_t(\kappa u)$ would be identically zero for all u , implying the image of L_κ lies in a proper $(k - 1)$ -dimensional subspace of $\mathbb{K}$, contradicting bijectivity. Similarly, a zero column would mean some nonzero basis vector maps to 0, contradicting injectivity. $\square$

6.4 The SCE output bit is an explicit XOR of evaluator bits

Fix $N \geq 2$, indices $i \in [N]$ and $t \in [k]$. We now define explicit $\mathbb{F}_2$ mask coefficients that represent $\text{SCE}_{N,i,t}(X)$ as an XOR of evaluator bits.

Definition 6.7 (Coefficient masks for a fixed output bit). *Fix $N \geq 2$, $i \in [N]$, and $t \in [k]$. For each $j \in [N]$, set*

$$\kappa_{i,j} := (C_N)_{i,j} \in \mathbb{K}, \quad M_{i,j} := M(\kappa_{i,j}) \in \mathbb{F}_2^{k \times k}.$$

Define the mask coefficients

$$m_{j,\ell}^{(N,i,t)} := (M_{i,j})_{t,\ell} \in \mathbb{F}_2 \quad \text{for } j \in [N], \ell \in [k].$$

When N, i, t are clear, we write $m_{j,\ell}$ for $m_{j,\ell}^{(N,i,t)}$.

Lemma 6.8 (Bit-linear form for SCE). *For every evaluator X ,*

$$\text{SCE}_{N,i,t}(X) = \bigoplus_{j \in [N]} \bigoplus_{\ell \in [k]} \left(m_{j,\ell}^{(N,i,t)} \cdot x_{j,\ell} \right),$$

where $x_{j,\ell} \in \mathbb{F}_2$ is the ℓ -th output bit of $X(j)$, and the product is the usual multiplication in $\mathbb{F}_2$.

Proof. By definition,

$$(C_N x(X))_i = \sum_{j \in [N]} (C_N)_{i,j} x_j(X) = \sum_{j \in [N]} \kappa_{i,j} x_j(X),$$

where the sum is in $\mathbb{K}$. Write $x_j(X) = \sum_{\ell=0}^{k-1} x_{j,\ell} \alpha^\ell$. By Lemma 6.5, the t -th coordinate of $\kappa_{i,j} x_j(X)$ equals

$$\pi_t(\kappa_{i,j} x_j(X)) = \bigoplus_{\ell \in [k]} (M(\kappa_{i,j}))_{t,\ell} x_{j,\ell} = \bigoplus_{\ell \in [k]} m_{j,\ell}^{(N,i,t)} x_{j,\ell}.$$

Taking π_t of the full sum and using that π_t is $\mathbb{F}_2$ -linear (it is a coordinate projection; Definition 2.2) yields

$$\text{SCE}_{N,i,t}(X) = \pi_t((C_N x(X))_i) = \bigoplus_{j \in [N]} \bigoplus_{\ell \in [k]} m_{j,\ell}^{(N,i,t)} x_{j,\ell}.$$

$\square$

Lemma 6.9 (Computability of a single mask bit). *Fix the field representation from Definition 2.1, and fix $N \geq 2$. Given indices $i, j \in [N]$ and $t, \ell \in [k]$, the coefficient bit $m_{j,\ell}^{(N,i,t)} \in \mathbb{F}_2$ can be computed in time polynomial in k .*

Proof. We describe an explicit procedure.

1. Compute $a_i = \text{enc}(i)$ and $b_j = \text{enc}(N + j)$ as k -bit coordinate vectors in the basis $\mathcal{B}$. This is direct from Definition 2.3.
2. Compute $d = a_i - b_j \in \mathbb{K}$. In characteristic two this is coordinate-wise XOR.
3. By Lemma 6.2, $d \neq 0$, hence invertible. Compute $d^{-1} \in \mathbb{K}$ using standard polynomial arithmetic in $\mathbb{F}_2[z]/(p_k(z))$; for example, compute the multiplicative inverse of $d(z)$ modulo $p_k(z)$ using the extended Euclidean algorithm for polynomials. This runs in time polynomial in k .
4. Set $\kappa_{i,j} = d^{-1} \in \mathbb{K}$.
5. To compute the matrix entry $M(\kappa_{i,j})_{t,\ell}$, compute the product $\kappa_{i,j} \cdot \alpha^\ell \in \mathbb{K}$ in the polynomial representation, reduce modulo p_k , and read off its t -th coordinate. By Lemma 6.5, the coordinate vector of $\kappa_{i,j} \alpha^\ell$ is exactly the ℓ -th column of $M(\kappa_{i,j})$. Thus the desired entry is its t -th bit.

Each step is polynomial in k , so the overall procedure is polynomial in k . $\square$

Remark 6.10 (Nonzero rows of mask matrices). *Since $\kappa_{i,j} = (C_N)_{i,j} \neq 0$, Lemma 6.6 implies that for each fixed $t \in [k]$ and each $j \in [N]$, there exists at least one $\ell \in [k]$ with $m_{j,\ell}^{(N,i,t)} = 1$. This will later be used in an information-theoretic lower bound for a restricted oracle model.*

7 Homogeneous $\mathbb{F}_2$ -Linear Systems and Bounded Gadgets

We now introduce the intermediate combinatorial object used throughout the remainder of the paper: bounded-arity, bounded-occurrence homogeneous linear systems over $\mathbb{F}_2$. We then develop two reusable gadgets—EqTree and XorTree—that allow us to copy a single bit to many locations and to aggregate large XORs while maintaining constant arity and occurrence.

7.1 Systems, arity, and occurrence

Definition 7.1 (Homogeneous $\mathbb{F}_2$ -linear system). *A homogeneous $\mathbb{F}_2$ -linear system is a pair*

$$\text{Sys} = (\mathcal{V}, \mathcal{E}),$$

where $\mathcal{V}$ is a finite set of Boolean variables (taking values in $\mathbb{F}_2$), and $\mathcal{E}$ is a finite set of linear equations over $\mathbb{F}_2$. Each equation $e \in \mathcal{E}$ is specified by a subset $S_e \subseteq \mathcal{V}$ and has the form

$$\bigoplus_{v \in S_e} v = 0.$$

A solution is an assignment $\sigma : \mathcal{V} \rightarrow \mathbb{F}_2$ satisfying all equations. A solution is nonzero if it is not identically zero, i.e., $\exists v \in \mathcal{V}$ with $\sigma(v) = 1$.

Definition 7.2 (Arity and occurrence). Let $\text{Sys} = (\mathcal{V}, \mathcal{E})$.

- The arity of an equation $e \in \mathcal{E}$ is $|S_e|$.
- The occurrence of a variable $v \in \mathcal{V}$ is the number of equations $e \in \mathcal{E}$ for which $v \in S_e$.

We say Sys has bounded arity if every equation has arity at most some absolute constant. We say Sys has bounded occurrence if every variable occurs in at most some absolute constant number of equations.

7.2 Equality gadget

Definition 7.3 (Equality constraint). For variables $u, v \in \mathcal{V}$, the equation

$$u \oplus v = 0$$

is called an equality constraint, since it enforces $u = v$.

We will use equality constraints organized in a tree to copy one root value to many leaves.

Definition 7.4 (Equality tree gadget: EqTree). Let $L \geq 1$. Let r be a distinguished variable (the root), and let $y_0, \dots, y_{L-1}$ be designated leaf variables. Choose a rooted binary tree $\mathcal{T}$ whose root is r and whose leaves are exactly $y_0, \dots, y_{L-1}$. Introduce fresh variables for the internal nodes of $\mathcal{T}$ (if any). For every edge (p, c) (parent p , child c) in $\mathcal{T}$, add the equality constraint

$$p \oplus c = 0.$$

The resulting set of constraints is denoted

$$\text{EqTree}(r; y_0, \dots, y_{L-1}).$$

Lemma 7.5 (Correctness of EqTree). In any solution of $\text{EqTree}(r; y_0, \dots, y_{L-1})$, every node in the tree (in particular, each leaf y_j) equals r .

Proof. Each edge constraint $p \oplus c = 0$ forces $p = c$. By induction on distance from the root, every node must equal the root value. $\square$

7.3 XOR-sum gadget

Definition 7.6 (XOR-sum tree gadget: XorTree). Let $L \geq 1$. Let S be a designated root variable, and let $z_0, \dots, z_{L-1}$ be designated leaf variables. Choose a rooted binary tree $\mathcal{T}$ whose root is S and whose leaves are exactly $z_0, \dots, z_{L-1}$. Introduce fresh variables for the internal nodes of $\mathcal{T}$ (if any). For every internal node s with children u and v , add the equation

$$s \oplus u \oplus v = 0.$$

Denote the resulting system by

$$\text{XorTree}(S; z_0, \dots, z_{L-1}).$$

Lemma 7.7 (Subtree XOR invariant for XorTree). Let $\mathcal{T}$ be the rooted binary tree underlying $\text{XorTree}(S; z_0, \dots, z_{L-1})$. For any node s in $\mathcal{T}$, let $\text{Leaves}(s)$ denote the set of leaf variables in the subtree rooted at s . In any solution σ of XorTree,

$$\sigma(s) = \bigoplus_{z \in \text{Leaves}(s)} \sigma(z).$$

In particular,

$$\sigma(S) = \bigoplus_{j=0}^{L-1} \sigma(z_j).$$

Proof. Proceed by induction on the height of node s . If s is a leaf z_j , then $\text{Leaves}(s) = \{z_j\}$ and the identity is trivial.

If s is an internal node with children u, v , the constraint $s \oplus u \oplus v = 0$ gives $\sigma(s) = \sigma(u) \oplus \sigma(v)$. By the induction hypothesis,

$$\sigma(u) = \bigoplus_{z \in \text{Leaves}(u)} \sigma(z), \quad \sigma(v) = \bigoplus_{z \in \text{Leaves}(v)} \sigma(z).$$

Since $\text{Leaves}(s) = \text{Leaves}(u) \cup \text{Leaves}(v)$ (disjoint union), we obtain the desired identity. The special case $s = S$ yields the root XOR formula. $\square$

7.4 Boundedness properties of the gadgets

Lemma 7.8 (Bounded arity and bounded occurrence of the gadgets). *Fix $L \geq 1$.*

1. *Every equation in $\text{EqTree}(r; y_0, \dots, y_{L-1})$ has arity 2.*
2. *Every equation in $\text{XorTree}(S; z_0, \dots, z_{L-1})$ has arity 3 (except for the degenerate case $L = 1$, where one may omit the gadget and set $S = z_0$).*
3. *If the rooted tree in EqTree is chosen to have maximum degree 3 (i.e., a rooted binary tree), then each variable in the gadget occurs in at most 3 equations.*
4. *In XorTree , each leaf z_j appears in exactly one equation (as a child of its parent), each internal node appears in exactly two equations (its own defining equation and its parent's equation), and the root S appears in exactly one equation. Hence every variable occurrence is at most 2 within the gadget.*

Proof. (1) and (2) follow directly from the definitions.

For (3), in a rooted binary tree each node has at most one parent edge and at most two child edges, so each node variable appears in at most $1 + 2 = 3$ constraints.

For (4), each internal node s participates in its own defining equation and (unless it is the root) in its parent's equation; leaves participate only as children, hence exactly once. $\square$

8 The System SysBit: Construction and Correctness

Fix $N \geq 2$, indices $i \in [N]$, $t \in [k]$, a target bit $b \in \mathbb{F}_2$, and an evaluator X . This section defines a homogeneous $\mathbb{F}_2$ -linear system

$$\text{SysBit}(N, i, t, b, X)$$

that has a nonzero solution if and only if $\text{SCE}_{N,i,t}(X) = b$.

Throughout this section, let $m_{j,\ell} = m_{j,\ell}^{(N,i,t)}$ be the mask coefficients from Definition 6.7, and let $x_{j,\ell}$ be the evaluator bits from Definition 6.3.

8.1 Intuition: a “switch” and a forced XOR check

We want to enforce

$$S = \bigoplus_{j \in [N]} \bigoplus_{\ell \in [k]} m_{j,\ell} x_{j,\ell},$$

and then check that $S = b$. Since our system must be homogeneous, we implement the check as

$$S \oplus (b \cdot \lambda_{\text{root}}) = 0,$$

where λ_{root} is a “switch” variable. The switch serves two purposes:

- It makes the system always satisfiable (the all-zero assignment).
- It ensures that nonzero satisfiability corresponds to the intended check: any nonzero solution must force $\lambda_{\text{root}} = 1$, and then the check becomes $S = b$.

8.2 Formal definition of SysBit

Definition 8.1 (The system $\text{SysBit}(N, i, t, b, X)$). *Define $\text{SysBit}(N, i, t, b, X) = (\mathcal{V}, \mathcal{E})$ as follows.*

Variables. *Include the following designated variables:*

- A root switch variable λ_{root} .
- Copy variables λ_j for each $j \in [N]$.
- Copy variables $\lambda_{j,\ell}$ for each $(j, \ell) \in [N] \times [k]$.
- Gating variables $u_{j,\ell}$ and $z_{j,\ell}$ for each $(j, \ell) \in [N] \times [k]$.
- Aggregation variables w_j for each $j \in [N]$.
- A final sum variable S .

In addition, include all internal variables introduced by the EqTree and XorTree gadgets instantiated below.

Equations. *Include the following constraints:*

(E1) *Copy λ_{root} to λ_j : include*

$$\text{EqTree}(\lambda_{\text{root}}; \lambda_0, \dots, \lambda_{N-1}).$$

(E2) *Copy λ_j to $\lambda_{j,\ell}$: for each $j \in [N]$, include*

$$\text{EqTree}(\lambda_j; \lambda_{j,0}, \dots, \lambda_{j,k-1}).$$

(E3) *Gate by evaluator bits $x_{j,\ell}$: for each $(j, \ell) \in [N] \times [k]$, include:*

- if $x_{j,\ell} = 0$, the unary equation $u_{j,\ell} = 0$;
- if $x_{j,\ell} = 1$, the binary equation $u_{j,\ell} \oplus \lambda_{j,\ell} = 0$.

(E4) *Gate by mask bits $m_{j,\ell}$: for each $(j, \ell) \in [N] \times [k]$, include:*

- if $m_{j,\ell} = 0$, the unary equation $z_{j,\ell} = 0$;
- if $m_{j,\ell} = 1$, the binary equation $z_{j,\ell} \oplus u_{j,\ell} = 0$.

(E5) Sum within each j : for each $j \in [N]$, include

$$\text{XorTree}(w_j; z_{j,0}, \dots, z_{j,k-1}).$$

(E6) Sum across j : include

$$\text{XorTree}(S; w_0, \dots, w_{N-1}).$$

(E7) Homogeneous check against b :

- if $b = 0$, include the unary equation $S = 0$;
- if $b = 1$, include the binary equation $S \oplus \lambda_{\text{root}} = 0$.

This completes the definition of $\text{SysBit}(N, i, t, b, X)$. (Each equation is of the form XOR-of-variables = 0, hence the system is homogeneous, and the all-zero assignment always satisfies all equations.)

8.3 The switch property

Lemma 8.2 (Switch lemma). *If σ is a solution of $\text{SysBit}(N, i, t, b, X)$ with $\sigma(\lambda_{\text{root}}) = 0$, then σ is the all-zero assignment. In particular, every nonzero solution must satisfy $\sigma(\lambda_{\text{root}}) = 1$.*

Proof. Assume $\sigma(\lambda_{\text{root}}) = 0$.

1. By (E1) and Lemma 7.5, $\sigma(\lambda_j) = \sigma(\lambda_{\text{root}}) = 0$ for all $j \in [N]$, and all internal variables of this EqTree gadget are 0.
2. Fix any j . By (E2) and Lemma 7.5, $\sigma(\lambda_{j,\ell}) = \sigma(\lambda_j) = 0$ for all $\ell \in [k]$, and the internal EqTree variables are 0.
3. Consider (E3). If $x_{j,\ell} = 0$, then $u_{j,\ell} = 0$. If $x_{j,\ell} = 1$, then the constraint $u_{j,\ell} \oplus \lambda_{j,\ell} = 0$ forces $u_{j,\ell} = \lambda_{j,\ell} = 0$. Thus $\sigma(u_{j,\ell}) = 0$ for all (j, ℓ) .
4. Consider (E4). If $m_{j,\ell} = 0$, then $z_{j,\ell} = 0$. If $m_{j,\ell} = 1$, then $z_{j,\ell} \oplus u_{j,\ell} = 0$ forces $z_{j,\ell} = u_{j,\ell} = 0$. Hence $\sigma(z_{j,\ell}) = 0$ for all (j, ℓ) .
5. Now (E5) says each w_j is the XOR of the $z_{j,\ell}$ (Lemma 7.7), so $\sigma(w_j) = 0$ for all j , and all internal XorTree nodes are 0.
6. Finally, (E6) implies S is the XOR of the w_j , so $\sigma(S) = 0$, and again internal nodes in this XorTree are 0.
7. The check (E7) is then satisfied automatically: if $b = 0$, it requires $S = 0$; if $b = 1$, it requires $S \oplus \lambda_{\text{root}} = 0$, which holds since both are 0.

Thus every designated variable is 0, and all gadget-internal variables are 0. Hence σ is the all-zero assignment. The final sentence follows immediately. $\square$

8.4 The forced XOR value when $\lambda_{\text{root}} = 1$

Lemma 8.3 (Forced sum lemma). *Let σ be a solution of $\text{SysBit}(N, i, t, b, X)$ with $\sigma(\lambda_{\text{root}}) = 1$. Then*

$$\sigma(S) = \bigoplus_{j \in [N]} \bigoplus_{\ell \in [k]} (m_{j,\ell} \cdot x_{j,\ell}).$$

Proof. Assume $\sigma(\lambda_{\text{root}}) = 1$.

1. By (E1) and Lemma 7.5, $\sigma(\lambda_j) = 1$ for all j . By (E2) and Lemma 7.5, $\sigma(\lambda_{j,\ell}) = 1$ for all j, ℓ .
2. Consider (E3). If $x_{j,\ell} = 0$, then $u_{j,\ell} = 0$. If $x_{j,\ell} = 1$, then $u_{j,\ell} \oplus \lambda_{j,\ell} = 0$ forces $u_{j,\ell} = 1$. Therefore,

$$\sigma(u_{j,\ell}) = x_{j,\ell}.$$

3. Consider (E4). If $m_{j,\ell} = 0$, then $z_{j,\ell} = 0$. If $m_{j,\ell} = 1$, then $z_{j,\ell} \oplus u_{j,\ell} = 0$ forces $z_{j,\ell} = u_{j,\ell} = x_{j,\ell}$. Hence,

$$\sigma(z_{j,\ell}) = m_{j,\ell} \cdot x_{j,\ell}.$$

4. By (E5) and Lemma 7.7, each w_j equals the XOR of $\{z_{j,\ell}\}_{\ell \in [k]}$, so

$$\sigma(w_j) = \bigoplus_{\ell \in [k]} \sigma(z_{j,\ell}) = \bigoplus_{\ell \in [k]} (m_{j,\ell} \cdot x_{j,\ell}).$$

5. By (E6) and Lemma 7.7 again,

$$\sigma(S) = \bigoplus_{j \in [N]} \sigma(w_j) = \bigoplus_{j \in [N]} \bigoplus_{\ell \in [k]} (m_{j,\ell} \cdot x_{j,\ell}).$$

□

8.5 Correctness: nonzero solution iff $\text{SCE} = b$

Theorem 8.4 (Correctness of SysBit). *$\text{SysBit}(N, i, t, b, X)$ has a nonzero solution if and only if $\text{SCE}_{N,i,t}(X) = b$.*

Proof. ($\Rightarrow$) Suppose $\text{SysBit}(N, i, t, b, X)$ has a nonzero solution σ . By Lemma 8.2, $\sigma(\lambda_{\text{root}}) = 1$. Then Lemma 8.3 gives

$$\sigma(S) = \bigoplus_{j,\ell} (m_{j,\ell} \cdot x_{j,\ell}).$$

By Lemma 6.8, the right-hand side equals $\text{SCE}_{N,i,t}(X)$. Finally, the check (E7) enforces $\sigma(S) = b$ when $\lambda_{\text{root}} = 1$:

- if $b = 0$, (E7) is $S = 0$;
- if $b = 1$, (E7) is $S \oplus \lambda_{\text{root}} = 0$, hence $S = \lambda_{\text{root}} = 1$.

Thus $\text{SCE}_{N,i,t}(X) = b$.

($\Leftarrow$) Suppose $\text{SCE}_{N,i,t}(X) = b$. We construct a nonzero solution σ .

- Set $\sigma(\lambda_{\text{root}}) = 1$, and for each EqTree in (E1)–(E2), set every variable in that tree to 1. All equality constraints are satisfied.

- For each (j, ℓ) , set $\sigma(u_{j,\ell}) = x_{j,\ell}$. Then (E3) is satisfied: if $x_{j,\ell} = 0$ it requires $u_{j,\ell} = 0$; if $x_{j,\ell} = 1$ it requires $u_{j,\ell} = \lambda_{j,\ell} = 1$.
- For each (j, ℓ) , set $\sigma(z_{j,\ell}) = m_{j,\ell} \cdot x_{j,\ell}$. Then (E4) is satisfied: if $m_{j,\ell} = 0$ it requires $z_{j,\ell} = 0$; if $m_{j,\ell} = 1$ it requires $z_{j,\ell} = u_{j,\ell}$.
- For each XorTree in (E5) and (E6), assign internal node values bottom-up so that each internal equation $s \oplus u \oplus v = 0$ holds; equivalently, set each internal node s to $\sigma(u) \oplus \sigma(v)$. Then the gadget constraints are satisfied and Lemma 7.7 holds. In particular,

$$\sigma(w_j) = \bigoplus_{\ell} \sigma(z_{j,\ell}), \quad \sigma(S) = \bigoplus_j \sigma(w_j) = \bigoplus_{j,\ell} m_{j,\ell} x_{j,\ell}.$$

By Lemma 6.8, the right-hand side equals $\text{SCE}_{N,i,t}(X) = b$.

- Finally, (E7) is satisfied because $\sigma(S) = b$ and $\sigma(\lambda_{\text{root}}) = 1$.

Thus σ is a nonzero solution. □

8.6 Bounded arity and bounded occurrence

Lemma 8.5 (Bounded arity and bounded occurrence of SysBit). *All equations in $\text{SysBit}(N, i, t, b, X)$ have arity at most 3. Moreover, there exists an absolute constant B (independent of N, i, t, b, X) such that every variable in SysBit occurs in at most B equations.*

Proof. Arity:

- EqTree constraints (E1)–(E2) are equalities of arity 2.
- XorTree constraints (E5)–(E6) have arity 3.
- The gating equations (E3)–(E4) have arity 1 or 2.
- The final check (E7) has arity 1 or 2.

Thus the maximum arity is 3.

Occurrence: choose all EqTree gadgets as rooted binary trees and all XorTree gadgets as rooted binary trees. Then:

- Within EqTree, each variable occurs in at most 3 equations (Lemma 7.8).
- Within XorTree, each variable occurs in at most 2 equations (Lemma 7.8).

Now check cross-gadget participation for each class of designated variables:

- Each λ_j participates in one EqTree in (E1) and one EqTree in (E2) and nowhere else.
- Each $\lambda_{j,\ell}$ participates in EqTree (E2) and in at most one gating equation in (E3).
- Each $u_{j,\ell}$ participates in at most one equation in (E3) and at most one equation in (E4).
- Each $z_{j,\ell}$ participates in at most one equation in (E4) and in exactly one XorTree equation (as a leaf) in (E5).
- Each w_j participates in one XorTree (E5) and one XorTree (E6).

- S participates in one XorTree (E6) and one check equation (E7).
- λ_{root} participates in EqTree (E1) and possibly in (E7) if $b = 1$.

Therefore each variable's total occurrence is bounded by a fixed constant (e.g., one may take $B = 8$), independent of the parameters. $\square$

Remark 8.6 (Size vs succinctness). *The system $\text{SysBit}(N, i, t, b, X)$ contains $\Theta(Nk)$ leaf-level variables and constraints. Later, we will represent such systems succinctly by local access, using the fact that each gating choice depends on either (i) evaluator bits $x_{j,\ell}$, obtainable from evaluating $X(j)$, or (ii) mask bits $m_{j,\ell}$, computable in $\text{poly}(k)$ time by Lemma 6.9. This “localizability” will be essential when translating SysBit into a succinct topological instance.*

9 A CW 2-Complex Encoding of a Bounded Linear System

In this part we define a purely topological encoding of bounded-arity, bounded-occurrence homogeneous systems over $\mathbb{F}_2$. Given such a system $\text{Sys} = (\mathcal{V}, \mathcal{E})$, we build a finite 2-dimensional CW complex

$$X := K^{\text{cw}}(\text{Sys})$$

with one 2-cell F_v per variable $v \in \mathcal{V}$ and a small “equation triangle” gadget per equation $e \in \mathcal{E}$. In the next section we prove that

$$H_2(K^{\text{cw}}(\text{Sys}); \mathbb{F}_2) \cong \text{Sol}(\text{Sys}),$$

i.e., the mod-2 second homology of the CW complex is naturally isomorphic to the solution space of the system.

9.1 Conventions for input systems

We work with homogeneous systems over $\mathbb{F}_2$. The reduction from SCE (Part II) produces systems of the following bounded form.

Convention 9.1 (Input systems for the CW encoding). *Fix a system $\text{Sys} = (\mathcal{V}, \mathcal{E})$. We assume:*

- $\mathcal{V}$ is a finite set of variables taking values in $\mathbb{F}_2$.
- $\mathcal{E}$ is a finite multiset of equations indexed by e . Each equation e is specified by a subset $S_e \subseteq \mathcal{V}$ of size $|S_e| = 3$ and encodes the homogeneous constraint

$$\bigoplus_{v \in S_e} v = 0.$$

- Each variable v has bounded occurrence: $r_v := |\text{Inc}(v)| \leq B$ for a fixed absolute constant B , where

$$\text{Inc}(v) := \{e \in \mathcal{E} : v \in S_e\}.$$

We also fix, for each v , an arbitrary but fixed ordering

$$\text{Inc}(v) = (e_{v,0}, e_{v,1}, \dots, e_{v,r_v-1}).$$

Definition 9.2 (Equation triangle gadget). *For an equation $e \in \mathcal{E}$, define three fresh vertices*

$$p(e, 0), p(e, 1), p(e, 2).$$

Add the three edges

$$\{p(e, 0), p(e, 1)\}, \{p(e, 1), p(e, 2)\}, \{p(e, 2), p(e, 0)\},$$

forming a triangle. We write the corresponding 1-cycle (over $\mathbb{F}_2$) as

$$\triangle_e := \{p(e, 0), p(e, 1)\} \oplus \{p(e, 1), p(e, 2)\} \oplus \{p(e, 2), p(e, 0)\}.$$

9.2 The CW 2-complex $K^{\text{cw}}(\text{Sys})$

We now define the main CW encoding. Intuitively, each equation e contributes an equation-triangle $\triangle_e$ in the 1-skeleton, and each variable v contributes a single 2-cell F_v whose attaching walk passes once around each $\triangle_e$ for $e \in \text{Inc}(v)$.

Definition 9.3 (CW complex encoding $K^{\text{cw}}(\text{Sys})$). *From $\text{Sys} = (\mathcal{V}, \mathcal{E})$, build a 2-dimensional CW complex*

$$X := K^{\text{cw}}(\text{Sys})$$

as follows.

- **0-cells.** *Create a vertex $b(v)$ for each $v \in \mathcal{V}$. For each equation $e \in \mathcal{E}$, create three vertices $p(e, 0), p(e, 1), p(e, 2)$. For each incidence $v \in S_e$, create a vertex $u(v, e)$. All these vertices are distinct.*
- **1-cells.**

– *For each equation $e \in \mathcal{E}$, include the three triangle edges*

$$\{p(e, 0), p(e, 1)\}, \{p(e, 1), p(e, 2)\}, \{p(e, 2), p(e, 0)\}.$$

– *For each incidence $v \in S_e$, include the two connector edges*

$$\{b(v), u(v, e)\}, \{u(v, e), p(e, 0)\}.$$

- **2-cells.** *For each variable $v \in \mathcal{V}$, attach a 2-cell F_v along a closed walk in the 1-skeleton that traverses, in order, the following “lollipop loop” for each incident equation $e_{v,j}$:*

$$b(v) \rightarrow u(v, e_{v,j}) \rightarrow p(e_{v,j}, 0) \rightarrow p(e_{v,j}, 1) \rightarrow p(e_{v,j}, 2) \rightarrow p(e_{v,j}, 0) \rightarrow u(v, e_{v,j}) \rightarrow b(v),$$

and then continues with $e_{v,j+1}$. Equivalently, the attaching map is the concatenation (in this fixed order) of these r_v loops.

Lemma 9.4 (Basic size bound). *Let $\text{Sys} = (\mathcal{V}, \mathcal{E})$ satisfy Convention 9.1, and let $X := K^{\text{cw}}(\text{Sys})$. Then the numbers of cells of X in dimensions 0, 1, 2 are each $O(|\mathcal{V}| + |\mathcal{E}|)$. More precisely:*

- $|X^{(0)}| = |\mathcal{V}| + 3|\mathcal{E}| + \sum_{v \in \mathcal{V}} r_v,$
- $|X^{(1)}| = 3|\mathcal{E}| + 2 \sum_{v \in \mathcal{V}} r_v,$
- $|X^{(2)}| = |\mathcal{V}|.$

Proof. By construction:

- **0-cells.** We add one vertex $b(v)$ for each $v \in \mathcal{V}$, three vertices $p(e, 0), p(e, 1), p(e, 2)$ for each $e \in \mathcal{E}$, and one vertex $u(v, e)$ for each incidence (v, e) with $v \in S_e$. The number of incidences is $\sum_{v \in \mathcal{V}} r_v$, giving the stated count.
- **1-cells.** Each equation contributes three triangle edges, and each incidence contributes two connector edges, giving $3|\mathcal{E}| + 2\sum_v r_v$.
- **2-cells.** We attach exactly one 2-cell F_v per variable v , giving $|\mathcal{V}|$.

Since the system has bounded arity $|S_e| = 3$, we have $\sum_v r_v = \sum_e |S_e| = 3|\mathcal{E}|$, so all three counts are $O(|\mathcal{V}| + |\mathcal{E}|)$. $\square$

Lemma 9.5 (Bounded local complexity (CW level)). *Let Sys satisfy Convention 9.1 with occurrence bound B , and let $X := K^{\text{cw}}(\text{Sys})$. Then:*

1. *Each 2-cell F_v is attached along a closed walk of length $7r_v \leq 7B$ in the 1-skeleton.*
2. *The maximum vertex degree in the 1-skeleton $X^{(1)}$ is at most $\max\{B, 5\}$.*

Proof. (1) By Definition 9.3, the attaching map of F_v is the concatenation of r_v lollipop loops. Each lollipop loop contains exactly 7 edge-steps, hence the walk length is $7r_v \leq 7B$.

(2) Consider the four types of vertices:

- A variable base vertex $b(v)$ is adjacent to $u(v, e)$ for each $e \in \text{Inc}(v)$, hence $\deg(b(v)) = r_v \leq B$.
- An incidence vertex $u(v, e)$ is adjacent only to $b(v)$ and $p(e, 0)$, hence $\deg(u(v, e)) = 2$.
- An equation vertex $p(e, 0)$ is adjacent to $p(e, 1)$ and $p(e, 2)$ (two triangle edges), and also to $u(v, e)$ for each $v \in S_e$. Since $|S_e| = 3$, we have $\deg(p(e, 0)) = 2 + |S_e| = 5$.
- The vertices $p(e, 1)$ and $p(e, 2)$ lie only on the triangle, so $\deg(p(e, 1)) = \deg(p(e, 2)) = 2$.

Therefore the maximum degree is at most $\max\{B, 5\}$. $\square$

10 Cellular Chains and the Isomorphism $H_2 \cong \text{Sol}(\text{Sys})$

Let $\text{Sys} = (\mathcal{V}, \mathcal{E})$ satisfy Convention 9.1, and let

$$X := K^{\text{cw}}(\text{Sys})$$

be the CW complex from Definition 9.3. In this section we compute $H_2(X; \mathbb{F}_2)$ in terms of the solution space of Sys .

10.1 Cellular chains for the CW encoding

Definition 10.1 (Cellular chain groups over $\mathbb{F}_2$). *Let $X = K^{\text{cw}}(\text{Sys})$ be as above. We consider cellular chains with coefficients in $\mathbb{F}_2$:*

$$\begin{aligned} C_2^{\text{cell}}(X; \mathbb{F}_2) &:= \mathbb{F}_2^{\{F_v : v \in \mathcal{V}\}}, \\ C_1^{\text{cell}}(X; \mathbb{F}_2) &:= \mathbb{F}_2^{E(X^{(1)})}, \\ C_0^{\text{cell}}(X; \mathbb{F}_2) &:= \mathbb{F}_2^{V(X)}. \end{aligned}$$

The boundary map

$$\partial_2^{\text{cell}} : C_2^{\text{cell}}(X; \mathbb{F}_2) \rightarrow C_1^{\text{cell}}(X; \mathbb{F}_2)$$

is defined by taking each 2-cell to the mod-2 sum of the (unoriented) 1-cells appearing in its attaching walk (with multiplicity mod 2). Since X is 2-dimensional, $\partial_3^{\text{cell}} = 0$, and therefore

$$H_2(X; \mathbb{F}_2) = \ker(\partial_2^{\text{cell}}).$$

The crucial observation is that each variable-cell F_v has boundary equal to the XOR of the equation triangles Δ_e for equations incident to v .

Lemma 10.2 (Boundary of F_v encodes incidences). *For each $v \in \mathcal{V}$,*

$$\partial_2^{\text{cell}}(F_v) = \bigoplus_{e \in \text{Inc}(v)} \Delta_e \quad \text{in } C_1^{\text{cell}}(X; \mathbb{F}_2),$$

where Δ_e is the equation-triangle cycle from Definition 9.2.

Proof. The attaching walk of F_v is a concatenation of lollipop loops, one for each $e \in \text{Inc}(v)$. Fix an incident equation e and consider its lollipop segment:

$$b(v) \rightarrow u(v, e) \rightarrow p(e, 0) \rightarrow p(e, 1) \rightarrow p(e, 2) \rightarrow p(e, 0) \rightarrow u(v, e) \rightarrow b(v).$$

In the cellular boundary over $\mathbb{F}_2$, this contributes all traversed edges, mod 2. The connector edge $\{b(v), u(v, e)\}$ is traversed twice (out and back), so it cancels. Likewise, the connector edge $\{u(v, e), p(e, 0)\}$ is traversed twice and cancels. The remaining edges in this segment are exactly the three triangle edges, contributing Δ_e . Summing over all $e \in \text{Inc}(v)$ gives the stated formula. $\square$

10.2 Solution space and the homology isomorphism

Definition 10.3 (Solution space). *The solution space of $\text{Sys} = (\mathcal{V}, \mathcal{E})$ is the $\mathbb{F}_2$ -vector space*

$$\text{Sol}(\text{Sys}) := \left\{ \sigma : \mathcal{V} \rightarrow \mathbb{F}_2 : \forall e \in \mathcal{E}, \bigoplus_{v \in S_e} \sigma(v) = 0 \right\}.$$

Definition 10.4 (Assignment-to-chain map Φ). *Define a linear map*

$$\Phi : \mathbb{F}_2^{\mathcal{V}} \rightarrow C_2^{\text{cell}}(X; \mathbb{F}_2)$$

by

$$\Phi(\sigma) := \bigoplus_{v \in \mathcal{V}: \sigma(v)=1} F_v.$$

In particular, for $\sigma \in \text{Sol}(\text{Sys})$, we view $\Phi(\sigma)$ as a 2-chain in X .

Lemma 10.5 (Boundary of $\Phi(\sigma)$). *For any assignment $\sigma : \mathcal{V} \rightarrow \mathbb{F}_2$,*

$$\partial_2^{\text{cell}}(\Phi(\sigma)) = \bigoplus_{e \in \mathcal{E}} \left(\bigoplus_{v \in S_e} \sigma(v) \right) \Delta_e.$$

Proof. By linearity,

$$\partial_2^{\text{cell}}(\Phi(\sigma)) = \bigoplus_{v:\sigma(v)=1} \partial_2^{\text{cell}}(F_v).$$

By Lemma 10.2, $\partial_2^{\text{cell}}(F_v) = \bigoplus_{e \in \text{Inc}(v)} \Delta_e$. Thus

$$\partial_2^{\text{cell}}(\Phi(\sigma)) = \bigoplus_{v:\sigma(v)=1} \bigoplus_{e \in \text{Inc}(v)} \Delta_e.$$

Re-indexing by equations, Δ_e appears once for each $v \in S_e$ with $\sigma(v) = 1$, hence its coefficient is $\bigoplus_{v \in S_e} \sigma(v)$. $\square$

Lemma 10.6 (Independence of equation triangles). *The set of 1-cycles $\{\Delta_e : e \in \mathcal{E}\}$ is linearly independent in $C_1^{\text{cell}}(X; \mathbb{F}_2)$.*

Proof. Consider any linear combination $\bigoplus_{e \in \mathcal{E}} c_e \Delta_e = 0$ in $C_1^{\text{cell}}(X; \mathbb{F}_2)$. For a fixed equation e , the edge $\{p(e, 0), p(e, 1)\}$ belongs only to Δ_e (since $p(e, 0), p(e, 1)$ are unique vertices for equation e and no other gadget uses that edge). Therefore, in the sum, the coefficient of $\{p(e, 0), p(e, 1)\}$ is exactly c_e , so $c_e = 0$. Since this holds for all e , all coefficients are zero, proving independence. $\square$

Theorem 10.7 (Second homology equals solution space). *Let Sys satisfy Convention 9.1 and let $X := K^{\text{cw}}(\text{Sys})$. Then*

$$H_2(X; \mathbb{F}_2) \cong \text{Sol}(\text{Sys})$$

as $\mathbb{F}_2$ -vector spaces.

Proof. Since X is 2-dimensional, Definition 10.1 gives

$$H_2(X; \mathbb{F}_2) = \ker(\partial_2^{\text{cell}}).$$

Every 2-chain is a unique $\mathbb{F}_2$ -linear combination of the cells $\{F_v\}_{v \in \mathcal{V}}$, hence is of the form $\Phi(\sigma)$ for a unique assignment $\sigma : \mathcal{V} \rightarrow \mathbb{F}_2$ (Definition 10.4).

By Lemma 10.5,

$$\partial_2^{\text{cell}}(\Phi(\sigma)) = \bigoplus_{e \in \mathcal{E}} \left(\bigoplus_{v \in S_e} \sigma(v) \right) \Delta_e.$$

By Lemma 10.6, the cycles $\{\Delta_e\}_{e \in \mathcal{E}}$ are linearly independent, so $\partial_2^{\text{cell}}(\Phi(\sigma)) = 0$ if and only if each coefficient $\bigoplus_{v \in S_e} \sigma(v)$ equals 0, i.e., if and only if $\sigma \in \text{Sol}(\text{Sys})$.

Therefore,

$$\ker(\partial_2^{\text{cell}}) = \Phi(\text{Sol}(\text{Sys})),$$

and Φ restricts to a linear isomorphism from $\text{Sol}(\text{Sys})$ onto $H_2(X; \mathbb{F}_2)$. $\square$

Corollary 10.8 (Betti-2 positivity criterion). *Let Sys satisfy Convention 9.1 and let $X := K^{\text{cw}}(\text{Sys})$. Then*

$$\beta_2(X; \mathbb{F}_2) > 0 \iff \text{Sys has a nonzero solution.}$$

Proof. By Theorem 10.7, $\beta_2(X; \mathbb{F}_2) = \dim H_2(X; \mathbb{F}_2) = \dim \text{Sol}(\text{Sys})$. Thus $\beta_2(X; \mathbb{F}_2) > 0$ iff $\text{Sol}(\text{Sys})$ contains a nonzero element, i.e., iff Sys has a nonzero solution. $\square$

10.3 Application to SysBit and SCE-Dec (CW level)

We now apply the CW encoding to the bounded-occurrence system $\text{SysBit}(N, i, t, b, X)$ built in Part II.

Corollary 10.9 (CW encoding of the SCE-Dec bit). *Fix parameters $N \geq 2$, $i \in [N]$, $t \in [k]$, $b \in \{0, 1\}$, and an evaluator X . Let*

$$X_{N,i,t,b,X}^{\text{cw}} := K^{\text{cw}}(\text{SysBit}(N, i, t, b, X)).$$

Then

$$\beta_2(X_{N,i,t,b,X}^{\text{cw}}; \mathbb{F}_2) > 0 \iff \text{SCE}_{N,i,t}(X) = b.$$

Proof. By Theorem 8.4 (Part II), $\text{SysBit}(N, i, t, b, X)$ has a nonzero solution if and only if $\text{SCE}_{N,i,t}(X) = b$. By Corollary 10.8, $\beta_2(K^{\text{cw}}(\text{Sys}); \mathbb{F}_2) > 0$ if and only if Sys has a nonzero solution. Combining these equivalences yields the claim. $\square$

Simplicialization, the ProbeBit Mapping, $\oplus \mathbf{P}$ -Completeness on $\text{Im}(\text{ProbeBit})$, $\text{SAT} \leq_{rp}$ via Isolation, and an Evaluation-Local Lower Bound

11 From $K^{\text{cw}}(\text{Sys})$ to a Bounded-Degree Simplicial 2-Complex

11.1 Why the simplicialization must respect boundary occurrences

In Part 3 we constructed, from a bounded-arity/occurrence homogeneous system $\text{Sys} = (\mathcal{V}, \mathcal{E})$, a CW 2-complex

$$X := K^{\text{cw}}(\text{Sys})$$

with one 2-cell F_v per variable $v \in \mathcal{V}$. The proof of Theorem 10.9 depended critically on the fact that each variable contributes a single 2-cell, so that the only 2-chains are $\mathbb{F}_2$ -linear combinations of these F_v .

A naïve “cone from a center vertex” simplicialization that identifies repeated boundary vertices would generally introduce extra 2-chains that can “select subsets of incident equation-cycles,” potentially enlarging H_2 and breaking the isomorphism $H_2 \cong \text{Sol}(\text{Sys})$. To avoid this, we must triangulate each 2-cell F_v using distinct boundary vertices per boundary occurrence before gluing to the 1-skeleton. The resulting structure is a triangular CW complex (a Δ -complex in the classical sense), and we then apply a barycentric subdivision to obtain an honest simplicial complex.

This section formalizes that two-step process and proves that it preserves geometric realization (hence homology).

11.2 The boundary walk for each CW 2-cell

Let $X = K^{\text{cw}}(\text{Sys})$ be as in Definition 9.3. Let $G := X^{(1)}$ denote its 1-skeleton (a finite graph).

For each $v \in \mathcal{V}$, the attaching map of the CW 2-cell F_v is, by construction (Definition 9.3), a closed walk in G . We fix a concrete encoding of that walk as a vertex sequence.

Definition 11.1 (Boundary walk encoding W_v). *For each $v \in \mathcal{V}$, let*

$$W_v := (w_0^{(v)}, w_1^{(v)}, \dots, w_{L_v}^{(v)})$$

be the vertex sequence of the attaching walk of F_v , with

$$w_{L_v}^{(v)} = w_0^{(v)}, \quad \{w_j^{(v)}, w_{j+1}^{(v)}\} \in E(G) \text{ for all } j \in \{0, \dots, L_v - 1\}.$$

(Thus L_v is the number of edge-steps in the closed walk.)

Remark 11.2 (Boundedness). *If Sys has variable occurrence $r_v := |\text{Inc}(v)| \leq B$, then in our CW encoding (Definition 9.3) the boundary walk is a concatenation of r_v “lollipop loops” each of length 7, so*

$$L_v \leq 7B.$$

This bound will be used later to prove bounded degree after subdivision.

11.3 Triangulating each 2-cell as a disk with distinct boundary occurrences

We now define, for each variable v , a simplicial disk T_v whose boundary cycle has exactly L_v distinct vertices, one for each boundary occurrence in W_v .

Definition 11.3 (The fan triangulation T_v of a polygonal disk). *Fix $v \in \mathcal{V}$ with boundary length $L_v \geq 1$. Define a simplicial 2-complex T_v as follows.*

- **Vertices:**

$$V(T_v) := \{c_v\} \cup \{q_j^{(v)} : j \in [L_v]\},$$

where all vertices are distinct and c_v is a designated “center.”

- **Edges:** *include all boundary edges $\{q_j^{(v)}, q_{j+1}^{(v)}\}$ for $j \in [L_v]$ (indices mod L_v), and all cone edges $\{c_v, q_j^{(v)}\}$ for $j \in [L_v]$.*
- **Triangles:** *for each $j \in [L_v]$, include the triangle*

$$\tau_j^{(v)} := \{c_v, q_j^{(v)}, q_{j+1}^{(v)}\}.$$

This is the standard fan triangulation of an L_v -gon from a center vertex.

Lemma 11.4 ($|T_v|$ is a topological disk). *The geometric realization $|T_v|$ is homeomorphic to the closed disk D^2 , and the subcomplex induced by $\{q_j^{(v)}\}_{j \in [L_v]}$ realizes as a simple cycle homeomorphic to S^1 .*

Proof. Choose a strictly convex polygon $P \subset \mathbb{R}^2$ with vertices $p_0, \dots, p_{L_v-1}$ in cyclic order, and choose a point $p_\star$ in the interior of P . Define a simplicial map $f : |T_v| \rightarrow P$ by sending

$$c_v \mapsto p_\star, \quad q_j^{(v)} \mapsto p_j,$$

and extending linearly on each triangle $\tau_j^{(v)}$. Because P is convex and each $\tau_j^{(v)}$ maps to the triangle $\text{conv}(p_\star, p_j, p_{j+1})$, these triangles form a triangulation of P . Hence f is a homeomorphism onto P , which is homeomorphic to D^2 . The boundary cycle maps to $\partial P \cong S^1$. $\square$

11.4 Gluing triangulated disks to the 1-skeleton: a triangular CW complex

We now describe how to glue each $|T_v|$ to the 1-skeleton graph G along the boundary walk W_v , without identifying distinct boundary occurrences.

Definition 11.5 (Boundary identification map γ_v). *Fix $v \in \mathcal{V}$. Define a continuous map*

$$\gamma_v : \partial|T_v| \rightarrow |G|$$

as follows. For each boundary vertex $q_j^{(v)}$ (for $j \in [L_v]$), set

$$\gamma_v(q_j^{(v)}) := w_j^{(v)} \in V(G),$$

and extend linearly on each boundary edge $\{q_j^{(v)}, q_{j+1}^{(v)}\}$ to the corresponding edge $\{w_j^{(v)}, w_{j+1}^{(v)}\}$ in G .

Definition 11.6 (Triangular subdivision complex $K^\Delta(\text{Sys})$). *Let*

$$Y := \left(|G| \sqcup \bigsqcup_{v \in \mathcal{V}} |T_v| \right) / \sim$$

where $\sim$ is the equivalence relation generated by

$$x \sim \gamma_v(x) \quad \text{for all } v \in \mathcal{V} \text{ and all } x \in \partial|T_v|.$$

Define $K^\Delta(\text{Sys})$ to be the resulting 2-dimensional cell complex, whose cells are:

- *the 0-cells $V(G)$ and the cone centers $\{c_v\}$,*
- *the 1-cells are the edges of G and the cone/boundary edges of each T_v ,*
- *the 2-cells are the triangles $\tau_j^{(v)}$.*

Remark 11.7 (What Y is and is not). *The object $K^\Delta(\text{Sys})$ is a triangular CW complex (a Δ -complex): it is built from triangles glued along edges, but it may fail to be a simplicial complex because distinct vertices of a triangle can be identified in Y after gluing (e.g., if the boundary walk repeats a vertex). We will fix this by barycentric subdivision in Section 11.6.*

11.5 Homology preservation: triangular subdivision does not change the space

We show that $K^\Delta(\text{Sys})$ has the same geometric realization (up to homeomorphism) as the original CW complex $K^{\text{cw}}(\text{Sys})$.

Lemma 11.8 (Boundary-fixing homeomorphism between $|T_v|$ and D^2). *For each v , there is a homeomorphism*

$$h_v : |T_v| \rightarrow D^2$$

such that h_v restricts to a homeomorphism $\partial|T_v| \rightarrow S^1$ taking the boundary cycle $q_0^{(v)}, q_1^{(v)}, \dots, q_{L_v-1}^{(v)}$ to L_v evenly spaced points on S^1 in cyclic order.

Proof. Fix a vertex v . By Lemma 11.4, $|T_v|$ is homeomorphic to D^2 , and the boundary subcomplex induced by

$\{q_j^{(v)}\}_{j \in [L_v]}$ realizes as a simple cycle homeomorphic to S^1 .
Let

$$g_v : |T_v| \rightarrow D^2$$

be any homeomorphism. Then g_v restricts to a homeomorphism

$$g_v|_{\partial|T_v|} : \partial|T_v| \rightarrow S^1.$$

Set $p_j := g_v(q_j^{(v)}) \in S^1$ for $j \in [L_v]$. Since $g_v|_{\partial|T_v|}$ is a homeomorphism, the points $p_0, p_1, \dots, p_{L_v-1}$ are distinct and occur on S^1 in the same cyclic order as the boundary cycle $q_0^{(v)}, q_1^{(v)}, \dots, q_{L_v-1}^{(v)}$.

Define target points on the unit circle

$$\zeta_j := e^{2\pi i j / L_v} \in S^1, \quad j \in [L_v].$$

Choose an orientation of S^1 and write each $p_j = e^{i\theta_j}$ with angles

$0 \leq \theta_0 < \theta_1 < \dots < \theta_{L_v-1} < 2\pi$ (after possibly rotating the indexing by a cyclic

shift; this does not change the statement). Define a continuous strictly increasing, piecewise-linear map

$$\varphi : [\theta_0, \theta_0 + 2\pi] \rightarrow [0, 2\pi]$$

by prescribing $\varphi(\theta_j) = 2\pi j / L_v$ for $j = 0, 1, \dots, L_v - 1$ and $\varphi(\theta_0 + 2\pi) = 2\pi$,

and extending linearly on each interval $[\theta_j, \theta_{j+1}]$ (with $\theta_{L_v} := \theta_0 + 2\pi$). This induces a homeomorphism $\rho : S^1 \rightarrow S^1$ given by

$$\rho(e^{i\theta}) := e^{i\varphi(\theta)}.$$

By construction, $\rho(p_j) = \zeta_j$ for all $j \in [L_v]$.

Now extend ρ to a homeomorphism $\Psi : D^2 \rightarrow D^2$ by the Alexander trick:

write each $x \in D^2$ as $x = r \cdot \omega$ with $r \in [0, 1]$ and $\omega \in S^1$ (with the convention $0 = 0 \cdot \omega$),

and set

$$\Psi(0) := 0, \quad \Psi(r \cdot \omega) := r \cdot \rho(\omega) \quad \text{for } r \in (0, 1], \omega \in S^1.$$

The map Ψ is continuous and bijective, with continuous inverse obtained by the same formula using ρ^{-1} ,

hence Ψ is a homeomorphism of D^2 extending ρ on S^1 .

Finally, define

$$h_v := \Psi \circ g_v : |T_v| \rightarrow D^2.$$

Then h_v is a homeomorphism and, for each $j \in [L_v]$,

$$h_v(q_j^{(v)}) = \Psi(g_v(q_j^{(v)})) = \Psi(p_j) = \rho(p_j) = \zeta_j = e^{2\pi i j / L_v},$$

as required. Moreover h_v restricts to a homeomorphism $\partial|T_v| \rightarrow S^1$ sending the boundary cycle to L_v evenly spaced points in cyclic order.

□

Lemma 11.9 (Homeomorphism of pushouts under boundary-fixing maps). *Let A be a Hausdorff space and let $f, g : S^1 \rightarrow A$ be continuous maps. Let*

$$X := (A \sqcup D^2) / \approx_f, \quad Y := (A \sqcup D^2) / \approx_g$$

be the pushouts obtained by identifying $x \in S^1 = \partial D^2$ with $f(x)$ (resp. $g(x)$). Suppose there exists a homeomorphism $\psi : D^2 \rightarrow D^2$ such that $\psi|_{S^1} : S^1 \rightarrow S^1$ satisfies

$$g = f \circ \psi|_{S^1}.$$

Then X and Y are homeomorphic.

Proof. Define $\Phi : A \sqcup D^2 \rightarrow A \sqcup D^2$ by $\Phi|_A = \text{id}_A$ and $\Phi|_{D^2} = \psi$. This is a homeomorphism. Moreover, if $x \in S^1$ then in Y we identify $x \sim g(x)$, while in X we identify $\psi(x) \sim f(\psi(x))$. The condition $g = f \circ \psi|_{S^1}$ ensures that Φ respects the equivalence relations and induces a well-defined homeomorphism between the quotients. $\square$

Theorem 11.10 (Triangular subdivision preserves the space). *For any bounded system Sys , the geometric realizations are homeomorphic:*

$$|K^\Delta(\text{Sys})| \cong |K^{\text{cw}}(\text{Sys})|.$$

Proof. In the CW complex $K^{\text{cw}}(\text{Sys})$, each variable cell F_v is attached by a map $f_v : S^1 \rightarrow |G|$ given by the closed walk W_v (Definition 9.3). In the triangular complex $K^\Delta(\text{Sys})$, the corresponding disk $|T_v|$ is attached along its boundary by γ_v (Definition 11.5). By Lemma 11.8, we can identify $|T_v|$ with D^2 via a boundary-fixing homeomorphism; then Lemma 11.9 shows that attaching $|T_v|$ via γ_v yields the same pushout space as attaching D^2 via f_v . Doing this for all v (finite gluing) yields the claimed homeomorphism. $\square$

11.6 Barycentric subdivision to obtain an honest simplicial 2-complex

To obtain an actual simplicial complex (rather than a Δ -complex), we take a barycentric subdivision.

Definition 11.11 (Barycentric subdivision $\text{sd}(Y)$ in dimension 2). *Let Y be a triangular CW complex: its 2-cells are triangles, and the attaching maps identify entire edges to edges. Define $\text{sd}(Y)$ as the simplicial complex whose vertices are barycenters of cells of Y :*

$$V(\text{sd}(Y)) := \{b_\sigma : \sigma \text{ is a cell of } Y, \dim \sigma \in \{0, 1, 2\}\}.$$

Include a simplex $\{b_{\sigma_0}, b_{\sigma_1}, b_{\sigma_2}\}$ whenever

$$\sigma_0 \subset \sigma_1 \subset \sigma_2$$

is a chain of cells in Y with $\dim \sigma_j = j$.

Lemma 11.12 (Barycentric subdivision preserves geometric realization). *For any triangular CW complex Y , the realizations are homeomorphic:*

$$|\text{sd}(Y)| \cong |Y|.$$

Proof. It suffices to define a homeomorphism on each triangular 2-cell that is compatible on shared faces. Let τ be a 2-cell of Y with vertices u_0, u_1, u_2 and edges e_{01}, e_{12}, e_{20} . The subdivision $\text{sd}(\tau)$ is the standard barycentric subdivision of a triangle: it has vertices $b_{u_i}, b_{e_{ij}}, b_\tau$ and triangles corresponding to chains $u_i \subset e_{ij} \subset \tau$. Map $|\text{sd}(\tau)|$ linearly onto $|\tau|$ by sending each barycentric vertex to the geometric barycenter of the corresponding face in $|\tau|$. This yields a homeomorphism on $|\tau|$, and the constructions agree on shared edges because the barycenters of shared faces match. Gluing over all 2-cells yields a global homeomorphism. $\square$

Definition 11.13 (Final simplicial complex $K(\text{Sys})$). *Define*

$$K(\text{Sys}) := \text{sd}\left(K^\Delta(\text{Sys})\right).$$

By Lemma 11.12, $|K(\text{Sys})| \cong |K^\Delta(\text{Sys})|$, and by Theorem 11.10, $|K^\Delta(\text{Sys})| \cong |K^{\text{cw}}(\text{Sys})|$. Therefore $|K(\text{Sys})| \cong |K^{\text{cw}}(\text{Sys})|$.

11.7 Homology consequences and Betti-2 positivity

Corollary 11.14 (Homology preservation from CW to simplicial). *For all q ,*

$$H_q(K(\text{Sys}); \mathbb{F}_2) \cong H_q(K^{\text{cw}}(\text{Sys}); \mathbb{F}_2).$$

In particular,

$$H_2(K(\text{Sys}); \mathbb{F}_2) \cong \text{Sol}(\text{Sys}) \quad \text{and} \quad \beta_2(K(\text{Sys}); \mathbb{F}_2) > 0 \iff \text{Sys has a nonzero solution.}$$

Proof. By Theorem 11.10 and Lemma 11.12, $|K(\text{Sys})| \cong |K^{\text{cw}}(\text{Sys})|$. Homology over $\mathbb{F}_2$ is invariant under homeomorphism, so H_q isomorphic for all q . The identification of H_2 with $\text{Sol}(\text{Sys})$ then follows from Theorem 10.9. The Betti-2 positivity statement follows from Corollary 10.10. $\square$

11.8 Bounded degree of $K(\text{Sys})$

Lemma 11.15 (Bounded degree). *Assume Sys has arity ≤ 3 and variable occurrence $\leq B$. Then there exists a constant $\Delta' = \Delta'(B)$ such that the simplicial 2-complex $K(\text{Sys})$ is Δ' -bounded-degree (in the sense of Definition 2.9).*

Proof. By Lemma 9.5, the CW complex $K^{\text{cw}}(\text{Sys})$ has bounded local complexity with bound depending only on B . The triangular subdivision $K^\Delta(\text{Sys})$ replaces each 2-cell by a disk triangulation with $L_v \leq 7B$ triangles; thus it also has bounded local complexity: each vertex is incident to only constantly many edges and triangles, and each edge is incident to only constantly many triangles (because equation edges have arity ≤ 3 , and boundary/cone edges are local to a single variable disk).

Now consider barycentric subdivision $K(\text{Sys}) = \text{sd}(K^\Delta(\text{Sys}))$. Each vertex of $K(\text{Sys})$ is the barycenter of a cell of $K^\Delta(\text{Sys})$ (0-, 1-, or 2-cell). A barycenter of a triangle is adjacent only to the barycenters of its three edges; a barycenter of an edge is adjacent to its two endpoint barycenters and to the barycenters of triangles incident to that edge; and a barycenter of a vertex is adjacent to barycenters of incident edges. All these incidences are bounded by the local complexity bound, hence by a constant depending only on B . Therefore vertex degree in the 1-skeleton is bounded, which implies bounded-degree in the sense of Definition 2.9 for a simplicial 2-complex. $\square$

12 ProbeBit: A Succinct Bounded-Degree Simplicial Complex Encoding an SCE Bit

We now apply Section 11 to the specific bounded system $\text{SysBit}(N, i, t, b, X)$ defined in Part 2.

12.1 Definition of ProbeBit

Definition 12.1 (ProbeBit mapping). *Given an SCE-Dec instance (N, i, t, b, X) , define:*

1. $\text{Sys} := \text{SysBit}(N, i, t, b, X)$ (Definition 8.1).
2. $K := K(\text{Sys})$ (Definition 11.13), a bounded-degree simplicial 2-complex.

Define $\text{ProbeBit}(N, i, t, b, X)$ to be a local-oracle description D (Definition 2.10) of K .

Recoverable-parameter convention. *For later promise-family statements, we fix a canonical compilation of the four oracle circuits in D such that the tuple (N, i, t, b) and a canonical gate-list encoding of X occur as a syntactically delimited prefix within the circuit encodings (as parsed under Appendix C).*

Consequently, given $\text{Enc}(D)$ one can recover (N, i, t, b, X) in time $\text{poly}(|\text{Enc}(D)|)$ by parsing the circuit descriptions and reading this designated prefix. (This convention does not change the semantics of K_D ; it only makes the parameters efficiently recoverable on inputs promised to be in $\text{Im}(\text{ProbeBit})$.)

12.2 Correctness of ProbeBit

Theorem 12.2 (ProbeBit correctness). *Let (N, i, t, b, X) be any SCE-Dec instance and let*

$$D := \text{ProbeBit}(N, i, t, b, X).$$

Then

$$\beta_2(K_D; \mathbb{F}_2) > 0 \iff \text{SCE}_{N,i,t}(X) = b.$$

Proof. By Theorem 8.4, $\text{SysBit}(N, i, t, b, X)$ has a nonzero solution iff $\text{SCE}_{N,i,t}(X) = b$. By Corollary 11.14 applied to $\text{Sys} = \text{SysBit}$, we have

$$\beta_2(K(\text{SysBit}(N, i, t, b, X)); \mathbb{F}_2) > 0 \iff \text{SysBit}(N, i, t, b, X) \text{ has a nonzero solution.}$$

By Definition 12.1, $K_D = K(\text{SysBit}(N, i, t, b, X))$. Combining these equivalences yields the claim. $\square$

12.3 Succinctness and efficient local-oracle access

We briefly justify that $\text{ProbeBit}(N, i, t, b, X)$ can be realized as a local-oracle description D of the target simplicial complex, with encoding size polynomial in $|X| + \log N$ and oracle evaluation time polynomial in $|X| + \log N$. Full bit-level details are deferred to Appendix C (as already announced in Part 1).

Lemma 12.3 (Efficient local-oracle evaluation for ProbeBit outputs). *Let (N, i, t, b, X) be any SCE-Dec instance and let $D := \text{ProbeBit}(N, i, t, b, X)$ (Definition 12.1). Then:*

1. $|D| = \text{poly}(|X| + \log N)$.

2. The oracles $\text{End}_D, \text{Vert}_D, \text{IncE}_D, \text{IncT}_D$ (Definition 2.10) are computable in time $\text{poly}(|X| + \log N)$ per query.

Proof sketch (construction with finite casework). We outline how D is encoded and why each oracle call is efficient.

Step 1: Define the encoding of D . We include a designated, syntactically delimited prefix encoding (N, i, t, b) together with a canonical gate-list encoding of the circuit X , plus fixed-size parameters describing the system-to-CW construction and the subdivision convention. This ensures a verifier can recover (N, i, t, b, X) from D in $\text{poly}(|D|)$ time as required by Definition 12.1.

Step 2: Oracles for the system $\text{SysBit}(N, i, t, b, X)$. We must be able to locally list (i) the variables and (ii) the equations, and to compute for a given incidence whether a variable appears in an equation and with what coefficient. In SysBit , the coefficient bits are derived from the mask values $m_{j,\ell}^{(N,i,t)}$ and from the circuit output bits $x_{j,\ell}$; by Lemma 6.9 we can compute the mask bits in $\text{poly}(\log N)$ time, and we can evaluate $X(j)$ in $\text{poly}(|X| + \log N)$ time.

Step 3: Oracles for the CW complex $K^{\text{cw}}(\text{SysBit})$. The CW encoding from Part 3 has bounded local complexity and uses fixed gadget templates; a local query to the CW adjacency data reduces to: computing constant-size incidence lists in SysBit plus fixed finite lookup in the gadget template.

Step 4: Oracles for K^Δ and barycentric subdivision. The triangular subdivision replaces each variable disk by a fan triangulation with $L_v \leq 7B$ triangles (Remark 11.2), and glues boundary occurrences according to the walk W_v . Because L_v is constant-bounded, local queries in K^Δ reduce to constant-time decoding of the relevant boundary position and constant-size adjacency lookups. The barycentric subdivision $\text{sd}(K^\Delta)$ introduces new vertices corresponding to cells in K^Δ ; edges/triangles correspond to chains $0 \subset 1 \subset 2$. Since the local incidence degrees in K^Δ are bounded by constants depending only on B , the barycentric complex has bounded degree as in Lemma 11.15.

All oracle outputs reduce to:

- decoding a cell ID,
- computing a constant-size list of incident lower-/higher-dimensional cells, and
- translating these to barycentric vertices.

Each such operation is $\text{poly}(\log N + |X|)$ because:

- evaluating $X(j)$ is $\text{poly}(|X| + \log N)$,
- computing mask bits $m_{j,\ell}^{(N,i,t)}$ is $\text{poly}(\log N)$ time by Lemma 6.9, and
- all additional computations are fixed finite arithmetic and pointer arithmetic on $O(\log N)$ -bit indices.

Finally, the description size is polynomial in $|X| + \log N$ because the oracles are uniform circuits that hardwire only (N, i, t, b) and the encoding of X , plus fixed gadget logic. This completes the succinctness argument. $\square$

Remark 12.4 (Scope note (no hidden claim)). *Lemma 12.3 is a “construction sketch with finite casework.” It is logically sufficient for the complexity reductions below, because it specifies what must be computable and gives the core reason (bounded local complexity and bounded boundary length) why each oracle evaluation is polynomial time in the description length. A fully enumerated bit-level encoding is deferred to Appendix C to avoid distracting from the homological spine.*

13 $\oplus\mathbf{P}$ -Completeness of Π_{β_2} on $\text{Im}(\text{ProbeBit})$

Let

$$\mathcal{I}_{\text{Probe}} := \text{Im}(\text{ProbeBit})$$

be the promise family of local-oracle descriptions produced by ProbeBit .

13.1 Membership: $\text{SCE-Dec} \in \oplus\mathbf{P}$

Lemma 13.1 ($\text{SCE-Dec} \in \oplus\mathbf{P}$). *SCE-Dec (Definition 3.5) lies in $\oplus\mathbf{P}$.*

Proof. Fix input (N, i, t, b, X) . By Lemma 6.8,

$$\text{SCE}_{N,i,t}(X) = \bigoplus_{j \in [N]} \bigoplus_{\ell \in [k]} (m_{j,\ell}^{(N,i,t)} \cdot x_{j,\ell}),$$

where $x_{j,\ell}$ is the ℓ -th output bit of $X(j)$, and $m_{j,\ell}$ is computable in $\text{poly}(\log N)$ time by Lemma 6.9.

Define a nondeterministic polynomial-time machine M that:

1. Nondeterministically guesses an integer $u \in \{0, \dots, 2^r - 1\}$ where $r := \lceil \log_2(Nk) \rceil$.
2. If $u \geq Nk$, reject.
3. Otherwise decode $u \mapsto (j, \ell) \in [N] \times [k]$.
4. Compute $x_{j,\ell}$ by evaluating $X(j)$ and extracting bit ℓ .
5. Compute $m_{j,\ell}^{(N,i,t)}$ by Lemma 6.9.
6. Accept iff $m_{j,\ell} \cdot x_{j,\ell} = 1$.

Then $\#\text{acc}_M(N, i, t, X) \bmod 2$ equals the XOR of all terms $m_{j,\ell} x_{j,\ell}$, hence equals $\text{SCE}_{N,i,t}(X)$.

To decide equality to b with parity acceptance:

- If $b = 1$, accept iff parity is 1, i.e., use M as is.
- If $b = 0$, flip parity by adding one additional always-accepting branch; then parity equals $\text{SCE}_{N,i,t}(X) \oplus 1$, which is 1 iff $\text{SCE}_{N,i,t}(X) = 0$.

Thus $(N, i, t, b, X) \in \text{SCE-Dec}$ iff the accepting-path count is odd, proving membership in $\oplus\mathbf{P}$. $\square$

13.2 Hardness: $\oplus\text{SAT} \leq_m \text{SCE-Dec}$

Lemma 13.2 ($\oplus\text{SAT} \leq_m \text{SCE-Dec}$). *There exists a deterministic polynomial-time many-one reduction from $\oplus\text{SAT}$ to SCE-Dec .*

Proof. Let $\phi(y_0, \dots, y_{n-1})$ be an input Boolean formula. Set

$$N := 2^n, \quad k := \lceil \log_2(4N) \rceil, \quad \mathbb{K} := \mathbb{F}_{2^k},$$

and choose any $i \in [N]$ and $t \in [k]$ (for concreteness $i = 0, t = 0$).

We define an evaluator circuit X such that $\text{SCE}_{N,i,t}(X)$ equals the parity of satisfying assignments of ϕ .

On input $j \in [N] = \{0, \dots, 2^n - 1\}$, the circuit X does:

1. Interpret j as a Boolean assignment $y \in \{0, 1\}^n$ via binary expansion.
2. Compute $s := \phi(y) \in \{0, 1\}$.
3. Compute the Cauchy coefficient $\kappa_{i,j} := (C_N)_{i,j} = (a_i - b_j)^{-1} \in \mathbb{K}$, which is well-defined and nonzero by Lemma 6.2. Compute $\kappa_{i,j}^{-1}$ in time $\text{poly}(k)$.
4. Output the field element

$$x_j := s \cdot \kappa_{i,j}^{-1} \cdot \alpha^t \in \mathbb{K},$$

encoded in the fixed basis $\mathcal{B}$. (If $s = 0$, output 0; if $s = 1$, output $\kappa_{i,j}^{-1} \alpha^t$.)

Now compute:

$$\begin{aligned} (C_N x(X))_i &= \sum_{j \in [N]} \kappa_{i,j} x_j \\ &= \sum_{j \in [N]} \kappa_{i,j} \cdot (s \cdot \kappa_{i,j}^{-1} \alpha^t) \\ &= \sum_{j \in [N]} s \cdot \alpha^t. \end{aligned}$$

Since $\text{char}(\mathbb{K}) = 2$, this sum equals

$$\left(\bigoplus_{j \in [N]} s \right) \cdot \alpha^t.$$

Applying π_t (Definition 2.2), and noting that $\pi_t(\alpha^t) = 1$ and $\pi_t(0) = 0$, we obtain

$$\text{SCE}_{N,i,t}(X) = \bigoplus_{j \in [N]} s = \# \text{SAT}(\phi) \bmod 2.$$

Therefore, $\oplus \text{SAT}$ reduces to deciding whether $\text{SCE}_{N,i,t}(X) = 1$. Output the instance $(N, i, t, 1, X)$.

This mapping is computable in polynomial time in $|\phi|$ because:

- $N = 2^n$ is represented in binary using $O(n)$ bits,
- X can evaluate ϕ , compute $\kappa_{i,j}$ and its inverse using $\text{poly}(k) = \text{poly}(n)$ field arithmetic, and output $k = \Theta(n)$ bits.

Thus we have a deterministic many-one reduction $\oplus \text{SAT} \leq_m \text{SCE-Dec}$. □

13.3 $\oplus \mathbf{P}$ -completeness of Π_{β_2} on $\text{Im}(\text{ProbeBit})$

Theorem 13.3 ($\oplus \mathbf{P}$ -completeness on the ProbeBit promise family). $\Pi_{\beta_2} \upharpoonright_{\mathcal{I}_{\text{Probe}}}$ is $\oplus \mathbf{P}$ -complete under deterministic many-one reductions.

Proof. Hardness. By the standard fact that $\oplus \text{SAT}$ is $\oplus \mathbf{P}$ -complete [2], it suffices to reduce $\oplus \text{SAT}$ to $\Pi_{\beta_2} \upharpoonright_{\mathcal{I}_{\text{Probe}}}$.

By Lemma 13.2, $\oplus \text{SAT} \leq_m \text{SCE-Dec}$. By Theorem 12.2, for any SCE-Dec instance (N, i, t, b, X) ,

$$(N, i, t, b, X) \in \text{SCE-Dec} \iff \Pi_{\beta_2}(\text{ProbeBit}(N, i, t, b, X)) = 1.$$

Therefore $\oplus \text{SAT} \leq_m \Pi_{\beta_2} \upharpoonright_{\mathcal{I}_{\text{Probe}}}$.

Membership. On inputs promised to lie in $\mathcal{I}_{\text{Probe}}$, the input D allows recovery of (N, i, t, b, X) via the designated circuit-encoding prefix (Definition 12.1). By Lemma 13.1, $\text{SCE-Dec} \in \oplus\mathbf{P}$, hence there is a parity-NP machine deciding whether $\text{SCE}_{N,i,t}(X) = b$. By Theorem 12.2, this equals $\Pi_{\beta_2}(D)$. Therefore $\Pi_{\beta_2} \upharpoonright_{\mathcal{I}_{\text{Probe}}} \in \oplus\mathbf{P}$.

Combining hardness and membership yields $\oplus\mathbf{P}$ -completeness. $\square$

Remark 13.4 (Scope note (promise)). *The theorem is restricted to the promise family $\mathcal{I}_{\text{Probe}} = \text{Im}(\text{ProbeBit})$. No claim is made about $\oplus\mathbf{P}$ -completeness on arbitrary local-oracle inputs.*

14 A One-Sided Randomized SAT Reduction on $\text{Im}(\text{ProbeBit})$

14.1 Isolation lemma (standard external result)

Lemma 14.1 (Valiant–Vazirani isolation lemma — standard). *There exists a randomized polynomial-time procedure $\mathcal{I}$ that, given a Boolean formula ϕ , outputs formulas $\psi_1, \dots, \psi_m$ for $m = \text{poly}(|\phi|)$ such that:*

1. *If ϕ is unsatisfiable, then every ψ_r is unsatisfiable.*
2. *If ϕ is satisfiable, then with probability at least $1/\text{poly}(|\phi|)$, at least one ψ_r has exactly one satisfying assignment.*

See Valiant and Vazirani [1].

14.2 $\text{SAT} \leq_{rp} \Pi_{\beta_2}$ on $\text{Im}(\text{ProbeBit})$

Recall our definition of a one-sided randomized many-one reduction $A \leq_{rp} B$ from Definition 2.16.

Theorem 14.2 (One-sided randomized many-one reduction $\text{SAT} \leq_{rp} \Pi_{\beta_2} \upharpoonright_{\mathcal{I}_{\text{Probe}}}$). *There exists a one-sided randomized many-one reduction*

$$\text{SAT} \leq_{rp} \Pi_{\beta_2} \upharpoonright_{\mathcal{I}_{\text{Probe}}}.$$

Consequently, if Π_{β_2} on $\mathcal{I}_{\text{Probe}}$ were solvable in deterministic polynomial time, then $\mathbf{NP} \subseteq \mathbf{RP}$.

Proof. Given input formula ϕ , run the isolation procedure (Lemma 14.1) to obtain $\psi_1, \dots, \psi_m$ with $m = \text{poly}(|\phi|)$.

For each ψ_r , apply the deterministic reduction from Lemma 13.2 to build an SCE-Dec instance $(N_r, i, t, 1, X_r)$ such that

$$\text{SCE}_{N_r, i, t}(X_r) = 1 \iff \#\text{SAT}(\psi_r) \equiv 1 \pmod{2}.$$

If ψ_r has exactly one satisfying assignment, then $\#\text{SAT}(\psi_r) \equiv 1 \pmod{2}$, hence the SCE bit equals 1, and by Theorem 12.2,

$$\beta_2(K_{\text{ProbeBit}(N_r, i, t, 1, X_r)}) > 0.$$

To make a many-one reduction (single output instance), define D to be a local-oracle description of the disjoint union

$$K := \bigsqcup_{r=1}^m K_r, \quad K_r := K_{\text{ProbeBit}(N_r, i, t, 1, X_r)}.$$

(Disjoint union preserves bounded degree and admits a local-oracle description by adding a component selector in the vertex/edge/triangle IDs; this is routine and does not affect any homology statements.)

Since homology over $\mathbb{F}_2$ is additive over disjoint unions,

$$\beta_2(K; \mathbb{F}_2) = \sum_{r=1}^m \beta_2(K_r; \mathbb{F}_2),$$

hence $\beta_2(K) > 0$ iff there exists r with $\beta_2(K_r) > 0$. Therefore:

- If ϕ is unsatisfiable, then each ψ_r is unsatisfiable, so each K_r has $\beta_2(K_r) = 0$, hence $\beta_2(K) = 0$. The reduction outputs a NO-instance with probability 1 (one-sided).
- If ϕ is satisfiable, then with probability at least $1/\text{poly}(|\phi|)$, some ψ_r has exactly one satisfying assignment, hence the corresponding K_r has $\beta_2(K_r) > 0$, so $\beta_2(K) > 0$. Thus the reduction outputs a YES-instance with probability at least $1/\text{poly}(|\phi|)$.

This is exactly a one-sided randomized many-one reduction $\text{SAT} \leq_{rp} \Pi_{\beta_2} \upharpoonright_{\mathcal{I}_{\text{Probe}}}$. $\square$

Remark 14.3 (Scope note (promise)). *This reduction targets Π_{β_2} only on the promise family $\mathcal{I}_{\text{Probe}}$ (and disjoint unions thereof, which remain within a straightforward promise closure of the family).*

15 An Unconditional Evaluation-Local Lower Bound for SCE

We now formalize the oracle model and prove an information-theoretic lower bound. This is unconditional (no complexity assumptions).

15.1 Model: evaluation-local algorithms

Definition 15.1 (Evaluation-local algorithm). *Fix parameters (N, i, t) . An algorithm $\mathcal{A}$ is evaluation-local if it receives as input (N, i, t) and oracle access to an evaluator $X : [N] \rightarrow \{0, 1\}^k$, and it may query the oracle on indices $j \in [N]$ to obtain $X(j)$. It must output a bit intended to equal $\text{SCE}_{N,i,t}(X)$.*

A deterministic evaluation-local algorithm makes at most q oracle queries if on every oracle X it queries $X(j)$ on at most q distinct indices j .

15.2 Lower bound $q \geq N$

Theorem 15.2 (Deterministic evaluation-local lower bound). *Fix (N, i, t) . Any deterministic evaluation-local algorithm that computes $\text{SCE}_{N,i,t}(X)$ for all evaluators X must make at least N oracle queries in the worst case.*

Proof. Let $\mathcal{A}$ be deterministic and suppose it makes at most $N - 1$ oracle queries. Consider any execution of $\mathcal{A}$ on some oracle; let $Q \subseteq [N]$ be the set of queried indices, with $|Q| \leq N - 1$. Choose an unqueried index

$$j^* \in [N] \setminus Q.$$

By Lemma 6.6 applied to $\kappa_{i,j^*} = (C_N)_{i,j^*} \neq 0$, the multiplication matrix $M(\kappa_{i,j^*})$ is invertible, hence its t -th row is nonzero. Therefore there exists $\ell^* \in [k]$ such that

$$m_{j^*, \ell^*}^{(N,i,t)} = 1$$

(Definition 6.7; this is exactly Remark 6.10).

Now define two evaluators X and X' as follows:

- For all queried indices $j \in Q$, set $X(j) = X'(j)$ (identical outputs).
- For all unqueried indices $j \notin Q$, set $X(j) = 0$ and $X'(j) = 0$, except at $j = j^*$, where:

$$X(j^*) = 0 \in \mathbb{F}_2^k, \quad X'(j^*) \text{ equals } 0 \text{ except } x'_{j^*, \ell^*} = 1.$$

Then $\mathcal{A}$ receives identical oracle answers on all its queries under X and X' , hence produces the same output on both.

However, by Lemma 6.8,

$$\begin{aligned} \text{SCE}_{N,i,t}(X) \oplus \text{SCE}_{N,i,t}(X') &= \bigoplus_{j,\ell} m_{j,\ell} \cdot (x_{j,\ell} \oplus x'_{j,\ell}) \\ &= m_{j^*, \ell^*} \cdot (0 \oplus 1) \\ &= 1, \end{aligned}$$

since all other bits agree, and $m_{j^*, \ell^*} = 1$. Thus $\text{SCE}_{N,i,t}(X) \neq \text{SCE}_{N,i,t}(X')$, contradicting correctness of $\mathcal{A}$ on both oracles.

Therefore any deterministic evaluation-local algorithm must query all N indices in the worst case. $\square$

16 Deterministic Witness-Expansion: $\beta_2(K_\phi) = \#\text{SAT}(\phi)$

This section presents a deterministic construction mapping a Boolean formula ϕ to a bounded-degree simplicial 2-complex K_ϕ given in the local-oracle model, such that

$$\beta_2(K_\phi; \mathbb{F}_2) = \#\text{SAT}(\phi).$$

In particular, this yields deterministic $\text{SAT} \leq_m \Pi_{\beta_2}$ and parsimonious $\#\text{SAT} \leq_m \text{Compute-}\beta_2$.

A key technical constraint is that the local-oracle description requires explicit values (n_V, n_E, n_T) . Hence the per-assignment gadget size must be independent of whether ϕ is satisfied (otherwise one would need to know $\#\text{SAT}(\phi)$ just to compute n_T). We therefore use equal-size gadgets: a “YES gadget” with $\beta_2 = 1$ and a “NO gadget” with $\beta_2 = 0$, but both with the same counts $(|V|, |E|, |T|)$.

16.1 Two equal-size gadget blocks

All homology in this section is over $\mathbb{F}_2$.

16.1.1 A sphere gadget (triangulated S^2)

Definition 16.1 (Tetrahedral sphere complex S). *Let $S = (V_S, E_S, T_S)$ be the simplicial 2-complex with:*

- $V_S := \{0, 1, 2, 3\}$,
- $E_S := \binom{V_S}{2}$ (all 6 edges),
- $T_S := \binom{V_S}{3}$ (all 4 triangles).

This is the boundary complex of a tetrahedron, hence a triangulation of S^2 .

16.1.2 A padding forest (no 2-faces)

Definition 16.2 (Forest padding F). *Let $F = (V_F, E_F, T_F)$ be the simplicial 2-complex with:*

- $V_F := \{4, 5, 6, 7, 8, 9, 10, 11\}$ (8 vertices),
- $E_F := \{\{4, 5\}, \{5, 6\}, \{6, 7\}, \{7, 8\}, \{8, 9\}, \{9, 10\}\}$ (a length-6 path on $\{4, \dots, 10\}$),
- $T_F := \emptyset$,

and vertex 11 isolated.

Clearly F is 1-dimensional (no triangles), hence $\beta_2(F) = 0$.

16.1.3 The YES-block and NO-block (same (V, E, T) sizes)

Definition 16.3 (YES-block $G^{(1)}$). *Define the YES-block*

$$G^{(1)} := S \sqcup F.$$

Then

$$|V(G^{(1)})| = 4 + 8 = 12, \quad |E(G^{(1)})| = 6 + 6 = 12, \quad |T(G^{(1)})| = 4 + 0 = 4.$$

Definition 16.4 (NO-block $G^{(0)}$). *Define the NO-block $G^{(0)}$ as the disjoint union of four disjoint filled triangles:*

- *Vertices:* $V(G^{(0)}) := \{0, 1, \dots, 11\}$.
- *Triangles:* $T(G^{(0)}) := \{\{0, 1, 2\}, \{3, 4, 5\}, \{6, 7, 8\}, \{9, 10, 11\}\}$.
- *Edges:* $E(G^{(0)})$ are the union of the 3 edges of each triangle.

Then $|V(G^{(0)})| = 12$, $|E(G^{(0)})| = 12$, $|T(G^{(0)})| = 4$.

Lemma 16.5 (Bounded degree and equal-size property). *Both $G^{(1)}$ and $G^{(0)}$ are bounded-degree simplicial 2-complexes with maximum vertex degree ≤ 3 . Moreover,*

$$(|V|, |E|, |T|) \text{ is the same for } G^{(1)} \text{ and } G^{(0)}, \text{ namely } (12, 12, 4).$$

Proof. Equal-size counts were verified in Definitions 16.3–16.4. For bounded degree:

- In S , every vertex is incident to exactly 3 edges and 3 triangles.
- In F , path vertices have degree ≤ 2 , and vertex 11 has degree 0; no triangles.
- In $G^{(0)}$, each triangle vertex is incident to 2 edges and 1 triangle.

Disjoint union does not increase local degrees. Thus the maximum degree is 3. □

16.2 Computing β_2 of the blocks

16.2.1 $\beta_2(S) = 1$

Lemma 16.6 ($\beta_2(S; \mathbb{F}_2) = 1$). *For the tetrahedral sphere complex S from Definition 16.1, $\beta_2(S; \mathbb{F}_2) = 1$.*

Proof. In S , we have $|T_S| = 4$ and $|E_S| = 6$, so $C_2(S) \cong \mathbb{F}_2^4$ and $C_1(S) \cong \mathbb{F}_2^6$. Since there are no 3-simplices, $H_2(S) = \ker(\partial_2)$.

Let the four triangles be

$$\tau_0 = \{0, 1, 2\}, \quad \tau_1 = \{0, 1, 3\}, \quad \tau_2 = \{0, 2, 3\}, \quad \tau_3 = \{1, 2, 3\}.$$

(1) Existence of a nonzero 2-cycle. Consider $c := \tau_0 \oplus \tau_1 \oplus \tau_2 \oplus \tau_3 \in C_2(S)$. Every edge of S lies in exactly two of the τ_i . Therefore in $\partial_2(c)$, each edge appears with coefficient $2 \equiv 0 \pmod{2}$, so $\partial_2(c) = 0$. Hence $c \in H_2(S)$, and $\dim H_2(S) \geq 1$.

(2) $\dim H_2(S) \leq 1$. It suffices to show $\text{rank}(\partial_2) \geq 3$, because then

$$\dim \ker(\partial_2) = \dim C_2 - \text{rank}(\partial_2) \leq 4 - 3 = 1.$$

Consider the boundaries of τ_0, τ_1, τ_2 . Each contains a unique edge not present in the other two:

- Edge $\{1, 2\}$ appears in $\partial_2(\tau_0)$ but not in $\partial_2(\tau_1)$ or $\partial_2(\tau_2)$.
- Edge $\{1, 3\}$ appears in $\partial_2(\tau_1)$ but not in $\partial_2(\tau_0)$ or $\partial_2(\tau_2)$.
- Edge $\{2, 3\}$ appears in $\partial_2(\tau_2)$ but not in $\partial_2(\tau_0)$ or $\partial_2(\tau_1)$.

Therefore $\partial_2(\tau_0), \partial_2(\tau_1), \partial_2(\tau_2)$ are linearly independent in $C_1(S)$, so $\text{rank}(\partial_2) \geq 3$.

Combining (1) and (2), we obtain $\dim H_2(S) = 1$, hence $\beta_2(S) = 1$. □

16.2.2 β_2 of triangles and forests

Lemma 16.7 (β_2 of a single filled triangle is 0). *Let D be the simplicial complex consisting of one triangle $\{a, b, c\}$ and its three edges and vertices. Then*

$$\beta_2(D; \mathbb{F}_2) = 0.$$

Proof. Here $C_2(D) \cong \mathbb{F}_2$ is spanned by $\{a, b, c\}$, and

$$\partial_2(\{a, b, c\}) = \{a, b\} \oplus \{a, c\} \oplus \{b, c\} \neq 0$$

in $C_1(D)$. Hence $\ker(\partial_2) = \{0\}$, so $H_2(D) = 0$ and $\beta_2(D) = 0$. □

Lemma 16.8 ($\beta_2(F; \mathbb{F}_2) = 0$). *For the forest padding F (Definition 16.2), $\beta_2(F) = 0$.*

Proof. F has no triangles, so $C_2(F) = 0$, hence $H_2(F) = 0$. □

Corollary 16.9 (β_2 of the blocks).

$$\beta_2(G^{(1)}; \mathbb{F}_2) = 1, \quad \beta_2(G^{(0)}; \mathbb{F}_2) = 0.$$

Proof. By Lemma 16.6 and Lemma 16.8, $\beta_2(S) = 1$ and $\beta_2(F) = 0$. By additivity over disjoint union (Lemma 16.10 below), $\beta_2(G^{(1)}) = \beta_2(S) + \beta_2(F) = 1$.

For $G^{(0)}$, it is the disjoint union of four single triangles, each with $\beta_2 = 0$ by Lemma 16.7; by additivity, $\beta_2(G^{(0)}) = 0$. □

16.3 Additivity of homology under disjoint unions

Lemma 16.10 (Homology of disjoint unions). *Let $K = K_1 \sqcup K_2$ be the disjoint union of simplicial 2-complexes. Then for each $q \in \{0, 1, 2\}$,*

$$H_q(K; \mathbb{F}_2) \cong H_q(K_1; \mathbb{F}_2) \oplus H_q(K_2; \mathbb{F}_2).$$

Proof. Since K_1 and K_2 are disjoint (no shared simplices), the chain groups split:

$$C_q(K) \cong C_q(K_1) \oplus C_q(K_2)$$

with respect to the basis of simplices. The boundary maps are block-diagonal:

$$\partial_q^K = \partial_q^{K_1} \oplus \partial_q^{K_2}.$$

Therefore $\ker(\partial_q^K) = \ker(\partial_q^{K_1}) \oplus \ker(\partial_q^{K_2})$ and similarly for images. Taking quotients yields the direct sum decomposition on homology. $\square$

Corollary 16.11 (Additivity of β_2). *For disjoint unions, $\beta_2(K_1 \sqcup K_2) = \beta_2(K_1) + \beta_2(K_2)$.*

16.4 The witness-expansion complex K_ϕ

Let ϕ be a Boolean formula on n variables $y_0, \dots, y_{n-1}$. We identify assignments with integers in $[2^n]$.

Definition 16.12 (Assignment decoding). *For $a \in [2^n]$, define $\text{asgn}(a) \in \{0, 1\}^n$ as the length- n binary expansion of a , i.e., $\text{asgn}(a) = (a_0, \dots, a_{n-1})$ with*

$$a = \sum_{j=0}^{n-1} a_j 2^j.$$

Definition 16.13 (Witness-expansion complex K_ϕ). *Let $M := 2^n$. For each $a \in [M]$, define a block*

$$\mathcal{C}_a := \begin{cases} G^{(1)} & \text{if } \phi(\text{asgn}(a)) = 1, \\ G^{(0)} & \text{if } \phi(\text{asgn}(a)) = 0. \end{cases}$$

Define

$$K_\phi := \bigsqcup_{a \in [M]} \mathcal{C}_a.$$

By Lemma 16.5, each $\mathcal{C}_a$ has bounded degree (at most 3), hence K_ϕ is also bounded-degree with the same constant bound.

16.5 Parsimonious equality $\beta_2(K_\phi) = \#\text{SAT}(\phi)$

Theorem 16.14 (Parsimonious equality). *For every Boolean formula ϕ ,*

$$\beta_2(K_\phi; \mathbb{F}_2) = \#\text{SAT}(\phi).$$

Proof. By Corollary 16.9, each block $\mathcal{C}_a$ contributes

$$\beta_2(\mathcal{C}_a) = \begin{cases} 1 & \text{if } \phi(\text{asgn}(a)) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

By additivity over disjoint unions (Corollary 16.11),

$$\beta_2(K_\phi) = \sum_{a \in [M]} \beta_2(\mathcal{C}_a) = \sum_{a \in [M]} \mathbf{1}[\phi(\text{asgn}(a)) = 1] = \#\text{SAT}(\phi).$$

□

Corollary 16.15 (Deterministic SAT reduction). *Let D_ϕ be a local-oracle description of K_ϕ . Then*

$$\phi \in \text{SAT} \iff \beta_2(K_{D_\phi}; \mathbb{F}_2) > 0 \iff \Pi_{\beta_2}(D_\phi) = 1.$$

Proof. ϕ is satisfiable iff $\#\text{SAT}(\phi) > 0$. By Theorem 16.14, $\#\text{SAT}(\phi) = \beta_2(K_\phi)$. Thus $\#\text{SAT}(\phi) > 0 \iff \beta_2(K_\phi) > 0$. □

Corollary 16.16 (Parsimonious $\#\text{SAT}$ reduction). *The mapping $\phi \mapsto K_\phi$ satisfies*

$$\#\text{SAT}(\phi) = \beta_2(K_\phi; \mathbb{F}_2).$$

Hence $\#\text{SAT} \leq_m \text{Compute-}\beta_2$.

16.6 Local-oracle description D_ϕ for K_ϕ

We now spell out a local-oracle description for K_ϕ consistent with Definition 2.10, including explicit global indexing.

16.6.1 Global sizes

Each block $\mathcal{C}_a$ has $(12, 12, 4)$ simplices in dimensions 0, 1, 2. Therefore:

Definition 16.17 (Global sizes for K_ϕ). *Let $M := 2^n$. Define*

$$n_V := 12M, \quad n_E := 12M, \quad n_T := 4M.$$

16.6.2 Global indexing scheme

Definition 16.18 (Component-local decoding). *For a global vertex index $v \in [n_V]$, define:*

$$a(v) := \left\lfloor \frac{v}{12} \right\rfloor \in [M], \quad r(v) := v \bmod 12 \in [12].$$

Similarly for global edge indices $e \in [n_E]$ and triangle indices $\tau \in [n_T]$,

$$a(e) := \left\lfloor \frac{e}{12} \right\rfloor, \quad s(e) := e \bmod 12, \quad a(\tau) := \left\lfloor \frac{\tau}{4} \right\rfloor, \quad q(\tau) := \tau \bmod 4.$$

Thus each vertex/edge/triangle index decodes to a component ID a and a local index within the 12-vertex, 12-edge, 4-triangle block.

16.6.3 Local tables for the two block types

Definition 16.19 (YES-block local edge endpoints and triangle vertices). *In $G^{(1)}$:*

- *Local edges $s \in [12]$ are:*
 - *Sphere edges (local $s = 0, \dots, 5$):*

$$0 : (0, 1), 1 : (0, 2), 2 : (0, 3), 3 : (1, 2), 4 : (1, 3), 5 : (2, 3).$$
 - *Forest edges (local $s = 6, \dots, 11$):*

$$6 : (4, 5), 7 : (5, 6), 8 : (6, 7), 9 : (7, 8), 10 : (8, 9), 11 : (9, 10).$$
- *Local triangles $q \in [4]$ are the sphere triangles:*

$$0 : (0, 1, 2), 1 : (0, 1, 3), 2 : (0, 2, 3), 3 : (1, 2, 3).$$

Definition 16.20 (NO-block local edge endpoints and triangle vertices). *In $G^{(0)}$, for each $q \in [4]$, the q -th triangle uses vertices $(3q, 3q + 1, 3q + 2)$. Define:*

- *Local triangles:*

$$q : (3q, 3q + 1, 3q + 2).$$
- *Local edges $s \in [12]$: write $q = \lfloor s/3 \rfloor$ and $r = s \bmod 3$. Then*

$$r = 0 : (3q, 3q + 1), \quad r = 1 : (3q, 3q + 2), \quad r = 2 : (3q + 1, 3q + 2).$$

16.6.4 Oracles

Let D_ϕ be the description consisting of (n_V, n_E, n_T) and circuits implementing the oracles below. We emphasize that the oracle computations use ϕ as a subroutine (compiled into a circuit) to decide whether block a is a YES-block or NO-block.

Definition 16.21 (Oracle End_{D_ϕ}). *Given edge index $e \in [n_E]$:*

1. *Decode $(a, s) = (a(e), s(e))$ (Definition 16.18).*
2. *Compute $b := \phi(\text{asgn}(a)) \in \{0, 1\}$ (Definition 16.12).*
3. *If $b = 1$, use Definition 16.19 to compute local endpoints $(u, v) \in [12]^2$. If $b = 0$, use Definition 16.20 to compute local endpoints $(u, v) \in [12]^2$.*
4. *Output global endpoints*

$$\text{End}_{D_\phi}(e) := (12a + u, 12a + v) \in [n_V]^2.$$

Definition 16.22 (Oracle Vert_{D_ϕ}). *Given triangle index $\tau \in [n_T]$:*

1. *Decode $(a, q) = (a(\tau), q(\tau))$.*
2. *Compute $b := \phi(\text{asgn}(a))$.*
3. *If $b = 1$, output local triangle vertices from Definition 16.19; if $b = 0$, output local triangle vertices from Definition 16.20.*
4. *Offset by $12a$ to get global vertices.*

Definition 16.23 (Fixed local incidence orders for $G^{(1)}$ and $G^{(0)}$). *Fix $\Delta := 3$.*

YES-block incidence ($b = 1$). Define $\text{incE}^{(1)} : [12] \times [3] \rightarrow ([12] \cup \{\perp\})$ and $\text{incT}^{(1)} : [12] \times [3] \rightarrow ([4] \cup \{\perp\})$ by the following tables. In each row, an entry (x_0, x_1, x_2) means $\text{inc}(r, 0) = x_0$, $\text{inc}(r, 1) = x_1$, $\text{inc}(r, 2) = x_2$.

r	$(\text{incE}^{(1)}(r, 0), \text{incE}^{(1)}(r, 1), \text{incE}^{(1)}(r, 2))$	$(\text{incT}^{(1)}(r, 0), \text{incT}^{(1)}(r, 1), \text{incT}^{(1)}(r, 2))$
0	(0, 1, 2)	(0, 1, 2)
1	(0, 3, 4)	(0, 1, 3)
2	(1, 3, 5)	(0, 2, 3)
3	(2, 4, 5)	(1, 2, 3)
4	(6, $\perp$, $\perp$)	($\perp$, $\perp$, $\perp$)
5	(6, 7, $\perp$)	($\perp$, $\perp$, $\perp$)
6	(7, 8, $\perp$)	($\perp$, $\perp$, $\perp$)
7	(8, 9, $\perp$)	($\perp$, $\perp$, $\perp$)
8	(9, 10, $\perp$)	($\perp$, $\perp$, $\perp$)
9	(10, 11, $\perp$)	($\perp$, $\perp$, $\perp$)
10	(11, $\perp$, $\perp$)	($\perp$, $\perp$, $\perp$)
11	($\perp$, $\perp$, $\perp$)	($\perp$, $\perp$, $\perp$)

NO-block incidence ($b = 0$). For $r \in [12]$, write $q := \lfloor r/3 \rfloor \in [4]$ and $t := r \bmod 3 \in \{0, 1, 2\}$. Define $\text{incE}^{(0)}(r, \ell)$ by:

$$(\text{incE}^{(0)}(r, 0), \text{incE}^{(0)}(r, 1), \text{incE}^{(0)}(r, 2)) = \begin{cases} (3q, 3q+1, \perp) & \text{if } t = 0, \\ (3q, 3q+2, \perp) & \text{if } t = 1, \\ (3q+1, 3q+2, \perp) & \text{if } t = 2, \end{cases}$$

and define $\text{incT}^{(0)}(r, \ell)$ by

$$\text{incT}^{(0)}(r, 0) = q, \quad \text{incT}^{(0)}(r, 1) = \text{incT}^{(0)}(r, 2) = \perp.$$

These orders are fixed and are consistent with the local edge/triangle tables in Definitions 16.19 and 16.20.

Definition 16.24 (Oracle IncE_{D_ϕ} and IncT_{D_ϕ}). Fix $\Delta := 3$. For a vertex index $v \in [n_V]$ and a slot $\ell \in [\Delta]$:

1. Decode $(a, r) = (a(v), r(v))$ (Definition 16.18), so $v = 12a + r$ with $r \in [12]$.
2. Compute the block type bit $b := \phi(\text{asgn}(a)) \in \{0, 1\}$ (Definition 16.12).
3. Compute local IDs

$$s := \text{incE}^{(b)}(r, \ell) \in [12] \cup \{\perp\}, \quad q := \text{incT}^{(b)}(r, \ell) \in [4] \cup \{\perp\},$$

using Definition 16.23.

4. Output global incidence values:

$$\text{IncE}_{D_\phi}(v, \ell) := \begin{cases} \perp & \text{if } s = \perp, \\ 12a + s & \text{if } s \in [12], \end{cases} \quad \text{IncT}_{D_\phi}(v, \ell) := \begin{cases} \perp & \text{if } q = \perp, \\ 4a + q & \text{if } q \in [4]. \end{cases}$$

Lemma 16.25 (Local incidence tables are consistent). *For each $b \in \{0, 1\}$ and each local vertex $r \in [12]$, the non- $\perp$ values among $\{\text{incE}^{(b)}(r, \ell) : \ell \in [3]\}$ enumerate exactly the local edges incident to r according to the local endpoint tables in Definitions 16.19 and 16.20. Likewise, the non- $\perp$ values among $\{\text{incT}^{(b)}(r, \ell) : \ell \in [3]\}$ enumerate exactly the local triangles incident to r according to the local triangle tables in the same definitions.*

Proof. For $b = 1$, this follows by direct inspection of the explicit YES tables in Definition 16.23 against the fixed local endpoints/triangles in Definition 16.19. For $b = 0$, each local block is a disjoint union of four filled triangles as in Definition 16.20, and the formulas in Definition 16.23 list exactly the two incident edges and the unique incident triangle per vertex. $\square$

Lemma 16.26 (Validity, bounded degree, and size of D_ϕ). *The description D_ϕ defined above satisfies $\text{Valid}(D_\phi)$ in the sense of Definition 2.12, and hence is a valid local-oracle description (Definition 2.10) of the simplicial 2-complex K_ϕ . Moreover:*

1. K_{D_ϕ} has bounded degree $\Delta = 3$.
2. The description length $|D_\phi|$ is polynomial in $|\phi|$.
3. Each oracle query runs in time $\text{poly}(|\phi|)$.

Proof. Validity. Within each component a , the endpoints/triangle-vertex data returned by End_{D_ϕ} and Vert_{D_ϕ} agree with the local tables (Definitions 16.19 and 16.20) by construction. The incidence lists returned by Definition 16.24 agree with those tables by Lemma 16.25. Therefore the range, loop-free/distinctness, soundness/completeness, bounded-degree (via $\Delta = 3$ slots and distinctness), and closure conditions in Definition 2.12 all hold, so $\text{Valid}(D_\phi)$ holds.

Bounded degree. Each block has maximum degree at most 3 and blocks are disjoint, hence the global degree bound is $\Delta = 3$.

Size/time. The paper's construction of D_ϕ is uniform and uses only fixed local tables together with a compiled representation of ϕ ; the resulting description size and per-query evaluation time are polynomial in $|\phi|$ as claimed in the main text. $\square$

16.7 Complexity consequences (deterministic, non-promise)

Theorem 16.27 (Deterministic many-one reduction $\text{SAT} \leq_m \Pi_{\beta_2}$). *There is a deterministic polynomial-time many-one reduction f mapping formulas ϕ to local-oracle descriptions D_ϕ such that*

$$\phi \in \text{SAT} \iff \Pi_{\beta_2}(D_\phi) = 1.$$

Proof. Take $f(\phi) := D_\phi$ as constructed above. By Corollary 16.15 and Lemma 16.26,

$$\phi \in \text{SAT} \iff \beta_2(K_{D_\phi}) > 0 \iff \Pi_{\beta_2}(D_\phi) = 1.$$

The mapping is polynomial-time because the output description size is polynomial in $|\phi|$. $\square$

Corollary 16.28 (Bitstring many-one reduction $\text{SAT} \leq_m L_{\beta_2}$). *Define the (total) mapping g on Boolean formulas by*

$$g(\phi) := \text{Enc}(D_\phi),$$

where D_ϕ is the local-oracle description constructed in the witness-expansion section. Then $\phi \in \text{SAT}$ if and only if $g(\phi) \in L_{\beta_2}$.

Proof. By Lemma 16.26 (as patched to reference Definition 2.12), $\text{Valid}(D_\phi)$ holds and $\text{Parse}(\text{Enc}(D_\phi)) = D_\phi \neq \perp$ by Appendix C. Hence $g(\phi) \in L_{\beta_2}$ iff $\beta_2(K_{D_\phi}; \mathbb{F}_2) > 0$. By Theorem 16.27, $\phi \in \text{SAT}$ iff $\Pi_{\beta_2}(D_\phi) = 1$, i.e. iff $\beta_2(K_{D_\phi}; \mathbb{F}_2) > 0$. $\square$

Theorem 16.29 (Parsimonious reduction $\# \text{SAT} \leq_m \text{Compute-}\beta_2$). *There is a deterministic polynomial-time many-one reduction from $\# \text{SAT}$ to $\text{Compute-}\beta_2$ satisfying*

$$\# \text{SAT}(\phi) = \beta_2(K_{D_\phi}; \mathbb{F}_2).$$

Proof. Immediate from Theorem 16.14 and Lemma 16.26. $\square$

Important limitation (explicitly stated). Theorem 16.27 shows NP-hardness of Π_{β_2} under deterministic many-one reductions in the local-oracle model. It does not establish that $\Pi_{\beta_2} \in \mathbf{NP}$ because, under succinct representations, certificates for $\beta_2 > 0$ (e.g., an explicit nonzero 2-cycle) may be exponentially large in $|D|$. On the other hand, Theorem 3.6 gives a baseline upper bound $\Pi_{\beta_2} \in \mathbf{PSPACE}$ (and likewise $L_{\beta_2} \in \mathbf{PSPACE}$ under our fixed parsing convention). Thus, while Π_{β_2} is NP-hard, its membership in $\mathbf{NP}$ (or other low-level classes) remains open in this succinct input model.

17 Unconditional Circuit Lower Bounds on an Explicit Parity-Based Subfamily

This section shows how to derive unconditional lower bounds against certain nonuniform circuit/formula classes by exhibiting an explicit parity-based restriction of Π_{β_2} that computes PARITY.

17.1 A bounded linear system for parity

Let $\text{PARITY} : \{0, 1\}^n \rightarrow \{0, 1\}$ be $\text{PARITY}(x) = x_0 \oplus \cdots \oplus x_{n-1}$.

We build, from x , a bounded-arity bounded-occurrence homogeneous system $\text{SysPar}(x)$ such that it has a nonzero solution iff $\text{PARITY}(x) = 1$.

Definition 17.1 (Parity system $\text{SysPar}(x)$). *Fix $n \geq 1$ and $x \in \{0, 1\}^n$. Define $\text{SysPar}(x)$ as a homogeneous $\mathbb{F}_2$ -linear system with variables:*

- a switch λ_{root} ,
- copies $\lambda_0, \dots, \lambda_{n-1}$,
- gated bits $u_0, \dots, u_{n-1}$,
- a sum variable S ,
- and internal variables of *EqTree* and *XorTree* gadgets.

Equations:

1. $\text{EqTree}(\lambda_{\text{root}}; \lambda_0, \dots, \lambda_{n-1})$.
2. For each $j \in [n]$, the gating equation:
 - if $x_j = 0$, include $u_j = 0$,
 - if $x_j = 1$, include $u_j \oplus \lambda_j = 0$.

3. $\text{XorTree}(S; u_0, \dots, u_{n-1})$.

4. *Final check* $S \oplus \lambda_{\text{root}} = 0$.

All equations have arity ≤ 3 . Variable occurrences are bounded by an absolute constant if the trees are chosen as rooted binary trees (as in Part 2).

Lemma 17.2 (Correctness of $\text{SysPar}(x)$). *$\text{SysPar}(x)$ has a nonzero solution if and only if $\text{PARITY}(x) = 1$.*

Proof. The proof is the same “switch” structure as Lemma 8.2 and Theorem 8.4, specialized to a 1-bit XOR.

- If $\lambda_{\text{root}} = 0$, then EqTree forces $\lambda_j = 0$ for all j . Then each gating equation forces $u_j = 0$ (either directly if $x_j = 0$, or because $u_j = \lambda_j = 0$ if $x_j = 1$). Then XorTree forces $S = \bigoplus_j u_j = 0$. The final check $S \oplus \lambda_{\text{root}} = 0$ holds. Hence the all-zero assignment is a solution.
- Consider a solution with $\lambda_{\text{root}} = 1$. EqTree forces $\lambda_j = 1$ for all j . Then each gating equation sets

$$u_j = \begin{cases} 0 & \text{if } x_j = 0, \\ 1 & \text{if } x_j = 1, \end{cases}$$

i.e., $u_j = x_j$. XorTree enforces $S = \bigoplus_j u_j = \bigoplus_j x_j = \text{PARITY}(x)$. The final check $S \oplus \lambda_{\text{root}} = 0$ becomes $S = 1$, hence requires $\text{PARITY}(x) = 1$.

Therefore, a nonzero solution exists (equivalently, a solution with $\lambda_{\text{root}} = 1$ exists) iff $\text{PARITY}(x) = 1$. □

17.2 Mapping parity systems to Π_{β_2} instances

We use the system-to-topology mapping already established:

- Part 3: $K^{\text{cw}}(\text{Sys})$ with $H_2 \cong \text{Sol}(\text{Sys})$.
- Part 4: simplicialization $K(\text{Sys})$ preserving H_2 .

Definition 17.3 (Parity-to- Π_{β_2} map $\Phi_{\oplus}$). *For $x \in \{0, 1\}^n$, define $\Phi_{\oplus}(x)$ to be a local-oracle description of the simplicial complex*

$$K(\text{SysPar}(x)).$$

(Concretely, $\Phi_{\oplus}(x)$ is obtained by applying the constructions of Sections 9–11 to $\text{SysPar}(x)$, then encoding the resulting bounded-degree simplicial complex by local-oracles.)

Theorem 17.4 (Parity realized by Π_{β_2} on an explicit subfamily). *For all $x \in \{0, 1\}^n$,*

$$\Pi_{\beta_2}(\Phi_{\oplus}(x)) = \text{PARITY}(x).$$

Proof. By Corollary 11.14, for any bounded system Sys ,

$$\beta_2(K(\text{Sys}); \mathbb{F}_2) > 0 \iff \text{Sys has a nonzero solution.}$$

Applying this to $\text{SysPar}(x)$, we get

$$\Pi_{\beta_2}(\Phi_{\oplus}(x)) = 1 \iff \text{SysPar}(x) \text{ has a nonzero solution.}$$

By Lemma 17.2, this is equivalent to $\text{PARITY}(x) = 1$. □

17.3 Projection/ $\mathbf{AC}^0$ nature of $\Phi_{\oplus}$: encoding-dependent and isolated

To translate Theorem 17.4 into nonuniform circuit lower bounds, we need a precise statement about how x is embedded into the bitstring encoding of $\Phi_{\oplus}(x)$. This is encoding-dependent, so we isolate it.

Definition 17.5 (Projection reduction; encoding-dependent). *Fix a concrete bit-level encoding scheme $\text{Enc}(\cdot)$ for local-oracle descriptions D . A family of maps*

$$\rho_n : \{0, 1\}^n \rightarrow \{0, 1\}^{m(n)}$$

is a projection if every output bit $(\rho_n(x))_k$ is either a constant in $\{0, 1\}$, or equals x_j , or equals $\neg x_j$, for some $j \in [n]$.

Lemma 17.6 (Projection property of $\Phi_{\oplus}$; proof in Appendix C). *Under the explicit encoding convention Enc fixed in Appendix C, the mapping*

$$x \mapsto \text{Enc}(\Phi_{\oplus}(x))$$

is a projection $\{0, 1\}^n \rightarrow \{0, 1\}^{m(n)}$ with $m(n) = \Theta(n)$.

Proof status: proved in Appendix C.

Comment: The construction of $\text{SysPar}(x)$ differs across x only in the local choice of which of two fixed gate patterns appears in each gating constraint; the encoding is chosen so that these choices correspond to designated bits equal to x_j (or $\neg x_j$), while all other description bits are fixed constants.

17.4 Unconditional lower bounds (via standard parity lower bounds)

We state three consequences. Each uses a standard lower bound for PARITY as a black box and therefore relies on a standard external lower bound (cited below).

17.4.1 $\mathbf{AC}^0$ lower bound

Theorem 17.7 (Unconditional $\mathbf{AC}^0$ lower bound for Π_{β_2} as a promise problem). *There is no nonuniform $\mathbf{AC}^0$ circuit family $\{C_m\}$ that correctly computes $\Pi_{\beta_2}(D)$ on all valid local-oracle descriptions D of bounded-degree simplicial 2-complexes (in the sense of Definition 2.10 and the chosen encoding).*

Proof. Assume for contradiction that such an $\mathbf{AC}^0$ family exists. Fix n . Consider the restriction of $C_{m(n)}$ to the set

$$\mathcal{F}_n := \{\text{Enc}(\Phi_{\oplus}(x)) : x \in \{0, 1\}^n\},$$

which consists entirely of valid instances by construction.

By Lemma 17.6, $\text{Enc}(\Phi_{\oplus}(x))$ is obtained from x by a projection ρ_n . Substituting the projection wires into $C_{m(n)}$ yields an $\mathbf{AC}^0$ circuit for $\text{PARITY}(x)$, because by Theorem 17.4 the output agrees with $\text{PARITY}(x)$ on all inputs x .

This contradicts the standard result $\text{PARITY} \notin \mathbf{AC}^0$ [3, 4]. Hence no such $\mathbf{AC}^0$ family exists. $\square$

17.4.2 De Morgan formula lower bound

Theorem 17.8 (Unconditional De Morgan formula lower bound for Π_{β_2} on an explicit subfamily). *Let $\mathcal{F}_n$ be as above. Any De Morgan formula that correctly computes Π_{β_2} on $\mathcal{F}_n$ must have size $\Omega(n^2)$.*

Proof. By Theorem 17.4 and Lemma 17.6, Π_{β_2} restricted to $\mathcal{F}_n$ computes PARITY under a projection. Projection substitutions do not increase formula size by more than a constant factor. Therefore any formula computing Π_{β_2} on $\mathcal{F}_n$ yields a formula computing PARITY on $\{0,1\}^n$ of comparable size. By the standard quadratic lower bound for parity in De Morgan formulas (e.g., Khrapchenko-type bounds; [7], the size must be $\Omega(n^2)$. $\square$

17.4.3 $\mathbf{AC}^0[p]$ lower bound for odd primes p

Theorem 17.9 (Unconditional $\mathbf{AC}^0[p]$ lower bound for odd primes). *Fix an odd prime p . There is no nonuniform $\mathbf{AC}^0[p]$ circuit family that correctly computes Π_{β_2} on all valid instances (in the same sense as Theorem 17.7). In particular, Π_{β_2} restricted to $\mathcal{F}_n$ is not in $\mathbf{AC}^0[p]$.*

Proof. The proof is identical to Theorem 17.7, replacing $\mathbf{AC}^0$ by $\mathbf{AC}^0[p]$ and using the standard result that $\text{PARITY} \notin \mathbf{AC}^0[p]$ for odd primes p (Razborov–Smolensky-type lower bounds [5, 6]. $\square$

Remark 17.10 (Scope note (explicit)). *Theorems 17.7–17.9 are unconditional but rely on standard external lower bounds for PARITY. They are statements about nonuniform circuit/formula families computing Π_{β_2} correctly on the promise set of valid encodings, which is the appropriate notion for a promise problem.*

18 Discussion, Limitations, and Open Issues

18.1 Summary of the validated contributions

The paper establishes the following validated core results:

1. **Algebraic linearization of SCE.** The SCE output bit $\text{SCE}_{N,i,t}(X)$ is an explicit $\mathbb{F}_2$ -linear form in evaluator bits (Part 2, Lemma 6.8).
2. **Bounded-occurrence system encoding.** The system $\text{SysBit}(N, i, t, b, X)$ has a nonzero solution iff $\text{SCE}_{N,i,t}(X) = b$ (Part 2, Theorem 8.4), with constant arity and occurrence (Lemma 8.5).
3. **Topological encoding of linear systems.** For a bounded system Sys , the CW complex $K^{\text{cw}}(\text{Sys})$ satisfies $H_2(K^{\text{cw}}(\text{Sys}); \mathbb{F}_2) \cong \text{Sol}(\text{Sys})$ (Part 3, Theorem 10.9). After a careful simplicialization via triangular subdivision and barycentric subdivision, homology is preserved (Part 4, Corollary 11.14).
4. **ProbeBit pipeline results (promise-family).** The map ProbeBit yields an equivalence $\beta_2 > 0 \iff \text{SCE} = b$ (Part 4, Theorem 12.2), leading to $\oplus \mathbf{P}$ -completeness and $\text{SAT} \leq_{\text{rp}}$ on the promise family $\text{Im}(\text{ProbeBit})$ (Part 4, Theorems 13.3 and 14.2), and a deterministic evaluation-local query lower bound $q \geq N$ for SCE (Part 4, Theorem 15.2).
5. **Witness-expansion deterministic reductions (non-promise).** The deterministic construction K_ϕ satisfies $\beta_2(K_\phi) = \#\text{SAT}(\phi)$, hence $\text{SAT} \leq_m \Pi_{\beta_2}$ and $\#\text{SAT} \leq_m \text{Compute-}\beta_2$ (Part 5, Theorems 16.27 and 16.29).

6. **Unconditional circuit lower bounds via parity subfamilies.** Using $\text{SysPar}(x)$ and the established system-to-topology mapping, we obtain an explicit parity-based restriction of Π_{β_2} and unconditional lower bounds against $\mathbf{AC}^0$, De Morgan formulas, and $\mathbf{AC}^0[p]$ (Part 5, Theorems 17.7–17.9), modulo standard parity lower bounds [3, 4, 7, 5, 6].

18.2 Why these results do not resolve $\mathbf{P}$ vs $\mathbf{NP}$

Theorem 16.27 shows NP-hardness of Π_{β_2} in the local-oracle model, and Theorem 13.3 gives $\oplus\mathbf{P}$ -completeness on a promise family. Neither implies $\mathbf{P} \neq \mathbf{NP}$, because:

- NP-hardness does not contradict $\mathbf{P} = \mathbf{NP}$; if $\mathbf{P} = \mathbf{NP}$, NP-hard problems can still be in $\mathbf{P}$.
- $\oplus\mathbf{P}$ -completeness on a promise family does not by itself yield a classical separation either.
- The paper does not establish $\Pi_{\beta_2} \in \mathbf{NP}$, which would be needed even to discuss NP-completeness in the usual total-function sense.

Thus, the results should be read as evidence of hardness phenomena for succinct topological invariants, not as a separation.

18.3 Promise-family hardness vs unrestricted deterministic hardness

There is a deliberate separation:

- ProbeBit results are strongest in parity-counting terms ($\oplus\mathbf{P}$ -completeness), but they are stated on $\text{Im}(\text{ProbeBit})$, a promise family.
- Witness-expansion yields an unrestricted deterministic reduction from SAT, but does not directly yield $\oplus\mathbf{P}$ -completeness statements.

18.4 Remaining technical dependencies and clearly identified gaps

This paper maintains strict marking of dependencies:

- **Standard external results (cited):**
 - Valiant–Vazirani isolation lemma [1] (used in Theorem 14.2).
 - $\oplus\text{SAT}$ is $\oplus\mathbf{P}$ -complete [2] (used in Theorem 13.3).
 - PARITY lower bounds for $\mathbf{AC}^0$, formulas, and $\mathbf{AC}^0[p]$ [3, 4, 7, 5, 6] (used in Theorems 17.7–17.9).
- **Encoding-dependent steps (proved or deferred):**
 - The “projection property” of the parity subfamily is explicitly proved in Appendix C (Lemma 17.6).
 - For ProbeBit, the fully explicit bit-level encoding of local-oracle circuits is nontrivial; Part 4’s Lemma 12.3 is a construction argument with finite casework, and full mechanical encoding details can be expanded further if a completely formal machine-checked implementation is required.

18.5 Open problems suggested by the results

1. **Sharper upper bounds for Π_{β_2} in the local-oracle model.** Theorem 3.6 gives a baseline bound $\Pi_{\beta_2} \in \mathbf{PSPACE}$. Determine whether substantially sharper upper bounds hold in this input model, e.g. whether Π_{β_2} lies in $\mathbf{NP}$, $\mathbf{coNP}$, Σ_2^P , or other low-level classes, and clarify what additional structural promises would be needed for NP-style witnesses.
2. **Promise elimination or promise robustness.** For the ProbeBit pipeline, identify natural syntactic constraints under which the promise family can be recognized (or made canonical), enabling sharper “non-promise” complexity statements.
3. **Derandomization of the $\text{SAT} \leq_{rp}$ route.** The isolation-based route is randomized. Deterministic analogues would require additional derandomization ingredients beyond this paper’s scope.
4. **Alternative invariants and dimensions.** Extend the framework to other homological invariants or to other dimensions, clarifying what aspects are specific to 2-complexes and mod-2 coefficients.

Appendix B. Boolean Circuits for $\mathbb{F}_{2^k}$ Arithmetic (Supplementary)

This appendix supports Remark 2.4 and Lemma 6.9 by giving explicit polynomial-size circuit constructions for basic $\mathbb{F}_{2^k}$ arithmetic under a fixed irreducible polynomial basis. The statements here are “internal completeness” results; they are not the main novelty of the paper.

Lemma 18.1 (Addition and multiplication in $\mathbb{F}_{2^k}$ have polynomial-size Boolean circuits). *Fix an irreducible polynomial $p_k(z) \in \mathbb{F}_2[z]$ of degree k and represent $\mathbb{F}_{2^k} \cong \mathbb{F}_2[z]/(p_k)$ using the polynomial basis $\{1, \alpha, \dots, \alpha^{k-1}\}$. Then:*

- *Addition is computed by k XOR gates (size $O(k)$).*
- *Multiplication is computed by a Boolean circuit of size $O(k^2)$.*

Proof. Addition is coordinate-wise XOR.

For multiplication, represent field elements as polynomials of degree $< k$ with coefficients in $\mathbb{F}_2$. Compute the polynomial product (degree $< 2k$) by convolution: each output coefficient is an XOR of ANDs of input bits; this uses $O(k^2)$ AND gates and $O(k^2)$ XOR gates. Then reduce modulo p_k : since p_k is fixed for the given k , reduction is an $\mathbb{F}_2$ -linear map from $\mathbb{F}_2^{2k-1}$ to $\mathbb{F}_2^k$, implemented by $O(k^2)$ XOR gates. Total size $O(k^2)$. $\square$

Lemma 18.2 (Inversion has polynomial-size circuits). *Under the same representation, inversion $u \mapsto u^{-1}$ on $\mathbb{F}_{2^k}^\times$ has polynomial-size Boolean circuits.*

Proof (explicit exponentiation). For nonzero u , $u^{-1} = u^{2^k-2}$ in $\mathbb{F}_{2^k}$. Compute u^{2^k-2} by repeated squaring and multiplication:

- Squaring in characteristic two is $\mathbb{F}_2$ -linear and can be implemented by a linear circuit of size $O(k^2)$ (or smaller).
- Use a standard square-and-multiply exponentiation strategy with $O(k)$ multiplications and squarings, giving total size polynomial in k when using Lemma 18.1 for multiplication.

This yields nonuniform polynomial-size circuits for inversion. $\square$

Appendix C: Fixed bit-level encoding of local-oracle descriptions

Integration note. This appendix replaces the previous sketch and fixes a single, explicit bit-level convention for Enc and Parse used throughout the paper.

C.1. A prefix-free self-delimiting integer code

For $t \in \mathbb{N}$, let $\ell(t) \geq 1$ be the bit-length of the standard binary representation of t (so $\ell(0) = 1$). Let $\text{bin}_{\ell(t)}^{\text{LSB}}(t) \in \{0, 1\}^{\ell(t)}$ be the $\ell(t)$ -bit little-endian encoding of t , i.e., $t = \sum_{i=0}^{\ell(t)-1} b_i 2^i$. Define the prefix-free codeword

$$\langle t \rangle := 1^{\ell(t)} 0 \parallel \text{bin}_{\ell(t)}^{\text{LSB}}(t).$$

Parsing $\langle t \rangle$ is deterministic: read a unary prefix $1^\ell 0$, then read ℓ payload bits.

C.2. Gate-list encoding of Boolean circuits (length-prefixed)

Fix the gate basis

$$\mathcal{G} = \{\text{CONST0}, \text{CONST1}, \text{NOT}, \text{AND}, \text{XOR}\}$$

with arities 0, 0, 1, 2, 2 and type IDs

$$\text{CONST0} \mapsto 0, \text{CONST1} \mapsto 1, \text{NOT} \mapsto 2, \text{AND} \mapsto 3, \text{XOR} \mapsto 4.$$

A (multi-output) Boolean circuit C is given syntactically by: an input length $k \in \mathbb{N}$, a gate list of length $g \in \mathbb{N}$ in topological order, and an output list of length $m \in \mathbb{N}$ selecting wires. Input wires are numbered $0, 1, \dots, k-1$ and gate $i \in \{0, 1, \dots, g-1\}$ outputs wire $k+i$. For gate i , each input wire index must lie in $\{0, 1, \dots, k+i-1\}$. Each output wire index must lie in $\{0, 1, \dots, k+g-1\}$.

Unprefixed payload encoding $\text{EncCircPayload}(C)$. Let C have parameters (k, m, g) . Define

$$\text{EncCircPayload}(C) := \langle k \rangle \langle m \rangle \langle g \rangle \parallel \left(\parallel_{i=0}^{g-1} (\langle \text{type}_i \rangle \parallel \parallel_{j=1}^{\text{arity}(\text{type}_i)} \langle \text{in}_{i,j} \rangle) \right) \parallel \left(\parallel_{q=0}^{m-1} \langle \text{out}_q \rangle \right),$$

where each $\text{type}_i \in \{0, 1, 2, 3, 4\}$, and $\text{in}_{i,j}, \text{out}_q$ are wire indices.

Length-prefixed encoding $\text{EncCirc}(C)$. Let $L := |\text{EncCircPayload}(C)|$ be the payload length in bits and define

$$\text{EncCirc}(C) := \langle L \rangle \parallel \text{EncCircPayload}(C).$$

This makes $\text{EncCirc}(C)$ self-delimiting inside concatenations.

Circuit parser ParseCirc . Given a bitstring y , define $\text{ParseCirc}(y)$ to be:

- $\perp$ if y cannot be written as $\langle L \rangle \parallel p \parallel y'$ with $|p| = L$,
- otherwise, parse p as $\text{EncCircPayload}(C)$; output $\perp$ if any parse fails, any type ID is outside $\{0, 1, 2, 3, 4\}$, any gate input violates the range bound, any output index is $\geq k+g$, or if there are leftover bits *inside* p ,
- if successful, output the pair (C, y') .

Thus EncCirc is injective and $\text{ParseCirc}(\text{EncCirc}(C) \parallel y') = (C, y')$.

C.3. Typed decoding conventions for oracle circuits

Fix the global degree bound constant $\Delta \geq 1$ of the local-oracle model. For $M \geq 0$, define the bit-width

$$w(M) := \lceil \log_2(M + 1) \rceil,$$

so $w(0) = 0$ is allowed (empty bitstrings). For $w \geq 0$, let $\text{bin}_w(t) \in \{0, 1\}^w$ be the w -bit little-endian encoding of t (padded with zeros), and let $\text{dec}_w : \{0, 1\}^w \rightarrow \{0, 1, \dots, 2^w - 1\}$ be the corresponding decode map.

Given (n_V, n_E, n_T) , set

$$w_V := w(n_V), \quad w_E := w(n_E), \quad w_T := w(n_T), \quad w_\Delta := w(\Delta).$$

Binary oracle-circuit interface (syntactic arities). A tag-0 description encodes four Boolean circuits with the following input/output lengths:

$$\begin{aligned} \text{End}^{\text{bin}} &: \{0, 1\}^{w_E} \rightarrow \{0, 1\}^{2w_V}, \\ \text{Vert}^{\text{bin}} &: \{0, 1\}^{w_T} \rightarrow \{0, 1\}^{3w_V}, \\ \text{IncE}^{\text{bin}} &: \{0, 1\}^{w_V + w_\Delta} \rightarrow \{0, 1\}^{1 + w_E}, \\ \text{IncT}^{\text{bin}} &: \{0, 1\}^{w_V + w_\Delta} \rightarrow \{0, 1\}^{1 + w_T}. \end{aligned}$$

Typed decoding (LSB-first split; $\perp$ by flag+payload). From these circuits, define typed oracles:

- For $e \in [n_E]$, evaluate $\text{End}^{\text{bin}}(\text{bin}_{w_E}(e)) \in \{0, 1\}^{2w_V}$, split into two blocks $\mathbf{u}, \mathbf{v} \in \{0, 1\}^{w_V}$, and set $\text{End}_D(e) := (\text{dec}_{w_V}(\mathbf{u}), \text{dec}_{w_V}(\mathbf{v}))$.
- For $\tau \in [n_T]$, evaluate $\text{Vert}^{\text{bin}}(\text{bin}_{w_T}(\tau)) \in \{0, 1\}^{3w_V}$, split into $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \{0, 1\}^{w_V}$, and set $\text{Vert}_D(\tau) := (\text{dec}_{w_V}(\mathbf{a}), \text{dec}_{w_V}(\mathbf{b}), \text{dec}_{w_V}(\mathbf{c}))$.
- For $(v, \ell) \in [n_V] \times [\Delta]$, evaluate $\text{IncE}^{\text{bin}}(\text{bin}_{w_V}(v) \parallel \text{bin}_{w_\Delta}(\ell)) \in \{0, 1\}^{1 + w_E}$. Write the first bit as a flag f and the remaining w_E bits as a payload $\mathbf{p}$. If $f = 0$ set $\text{IncE}_D(v, \ell) := \perp$; if $f = 1$ set $\text{IncE}_D(v, \ell) := \text{dec}_{w_E}(\mathbf{p})$.
- Define $\text{IncT}_D(v, \ell)$ analogously from IncT^{bin} (flag plus w_T -bit payload).

Range membership in $[n_V], [n_E], [n_T]$ is *not* enforced syntactically here; it is enforced semantically by $\text{Valid}(D)$ (Definition 2.12 in the main text).

C.4. Encoding $\text{Enc}(D)$ and total parsing Parse

Let Desc be the disjoint union of: (i) tag-0 objects of the form $(n_V, n_E, n_T, \text{End}^{\text{bin}}, \text{Vert}^{\text{bin}}, \text{IncE}^{\text{bin}}, \text{IncT}^{\text{bin}})$, and (ii) tag-1 objects of the form $\Phi_\oplus(x)$ from Section 17.

Tag 0 (generic encoding). For a tag-0 description D ,

$$\text{Enc}(D) := 0 \parallel \langle n_V \rangle \langle n_E \rangle \langle n_T \rangle \parallel \text{EncCirc}(\text{End}^{\text{bin}}) \parallel \text{EncCirc}(\text{Vert}^{\text{bin}}) \parallel \text{EncCirc}(\text{IncE}^{\text{bin}}) \parallel \text{EncCirc}(\text{IncT}^{\text{bin}}).$$

Tag 1 (template encoding for $\Phi_\oplus$). For $\Phi_\oplus(x)$ with $x \in \{0, 1\}^n$, define

$$\text{Enc}(\Phi_\oplus(x)) := 1 \parallel \langle n \rangle \parallel x_0 x_1 \cdots x_{n-1}.$$

For fixed n , this map is a projection (Lemma 17.6).

Total parser Parse. On input $x \in \{0, 1\}^*$:

- If x is empty, output $\perp$.
- Read the first tag bit.
- If the tag is 0, parse $\langle n_V \rangle, \langle n_E \rangle, \langle n_T \rangle$, compute w_V, w_E, w_T, w_Δ , then parse four circuits in sequence by repeatedly applying ParseCirc. Reject (output $\perp$) if any parse fails, if any circuit's declared input/output lengths do not match the arities in C.3, or if there are leftover bits after the fourth circuit. Otherwise output the resulting tag-0 description.
- If the tag is 1, parse $\langle n \rangle$, then read exactly n bits as $(x_0, \dots, x_{n-1})$ and reject if there are leftover bits. Otherwise output the tag-1 object $\Phi_\oplus(x)$.

Thus Parse is deterministic and total, and Enc is injective with $\text{Parse}(\text{Enc}(D)) = D$ for all $D \in \text{Desc}$.

C.5. Projection property for the parity-family encoding

We use the main-text definition of a projection (Definition 17.5).

Proof of Lemma 17.6. Fix n and consider the map $x \mapsto \text{Enc}(\Phi_\oplus(x)) = 1\|\langle n \rangle\|x$ from C.4. All bits in the prefix $1\|\langle n \rangle$ are constants depending only on n . The subsequent n output bits are exactly the input bits $x_0, \dots, x_{n-1}$ in order. Hence each output bit is either a constant or a literal copy of one input bit, i.e. a projection. Moreover $|\text{Enc}(\Phi_\oplus(x))| = \Theta(n)$. $\square$

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